The Conjugate Locus of the Rotating Geodesics in the Sub-Riemannian Problem on SE(2)

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Abstract

In the standard left-invariant sub-Riemannian problem on the group of plane motions the conjugate times of a rotating geodesic are known: there is one on each of two brackets in every Maxwell cell $[p_n^1, p_{n+1}^1)$, $n \geq 1$, the first conjugate time being the one on the left bracket of cell 1. These are two distinct times except at $k = k_0$, $\psi \equiv K \pmod{2K}$, where they coincide at $2nK$. We describe the image of these points under the exponential map, the conjugate locus. On every open sheet the map has corank one with a kernel tangent to the level sets of time, and the fold quantity along that kernel equals, up to an explicit nonzero factor, one polynomial Q in the Riccati variables of the first-time problem; a positive rational certificate for Q makes the sheet a fold surface. Its Maxwell edge and, below the critical modulus k_0, its half-period edge are cusp edges; the common point at k_0 has corank two and is a hyperbolic umbilic D_4^+; above k_0 the edge on which the two phase branches meet is a fold. The caustic is parametrized by the midpoint amplitude and the modulus in an explicit half-angle chart in which each of its eight open patches is embedded, the patches meet only along common source seams, the small-modulus limit is the identity point, and the separatrix limit leaves every compact set. The proofs use the first variation of the endpoint map, addition theorems, and the Riccati identities of the conjugate-times paper. The same identity governs every later sheet: each open sheet is a fold surface; for $k \neq k_0$ the four seams of a cell are a cusp edge at the Maxwell time, folds at the zeros of β_1 and α_1, and a cusp edge at the half period. At $k = k_0$ the α_1 and half-period seams meet in a hyperbolic umbilic; for large period count the fold quantity has an explicit leading term of order $1/n$ whose sign is that of $-\operatorname{sn} p \operatorname{cn} p$. Which sheets meet is restricted by an octant lemma and characterized through the scalar level function of the inverse problem: a target lies on the left sheet of cell n or the right sheet of cell j exactly when that function has a critical point at level $n-1$ or j, with the residual argument above or below the half period; the incidence map of the first four cells is computed numerically, the modulus along the chart curve is proved unimodal. A conditional sign criterion implies at most one critical point at levels ≥ 1 and at most two solutions per positive integer level on each admissible interval. That criterion is verified by interval arithmetic on the endpoint domain outside five declared exclusion sets (2.56% of the ambient volume; the fifth is a low-modulus wedge on which the interval engine does not close); the global count and the types of the two maxima remain conjectural. Exact identities and selected singular values, kernels and edge derivatives have symbolic and high-precision checks; Section 11 records their finite-sample scope and replay limitations.

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1 Introduction

The standard sub-Riemannian problem on SE(2) is

$$\dot{x} = u_1 \cos\theta, \qquad \dot{y} = u_1 \sin\theta, \qquad \dot{\theta} = u_2, \qquad \int_0^T (u_1^2 + u_2^2)\, dt \to \min, \tag{1}$$

with the identity as the initial point. Its normal extremals are parametrized by Jacobi elliptic functions, the pendulum structure of the vertical part giving five strata $C_1, \dots, C_5$ [12]; the conjugate times, the cut time and the optimal synthesis are known [14, 15], and on the rotating stratum C_2 the first conjugate time, bracketed by Sachkov, was made exact in [10]: for every modulus $0 < k < 1$ and every initial phase it is the unique zero of Sachkov's reduced Jacobian J_2 in the bracket $[p_0, p_1]$, and it is reached by inverting one of two elliptic branches.

This paper describes the first conjugate locus itself: the surface $\Sigma_1 \subset \mathrm{SE}(2)$ swept by the first conjugate points of the rotating geodesics, as the image of a smooth sheet in the source under the exponential map. Its singularities are the singularities of a Lagrangian map $\mathbb{R}^3 \to \mathbb{R}^3$, so the expected local types are folds (A_1), cusp edges (A_2), swallowtails (A_3) and isolated umbilic points ($D_4^{\pm}$) [3]. We use the map-germ indices A_1 (fold), A_2 (cusp), A_3 (swallowtail) and $D_4^{\pm}$ (umbilics) [18, 6, 3]; the first sheet contains no swallowtail, since Theorems 5.1 and 6.1 leave only folds and cusps on it. For the umbilics we follow Arnold's sign: D_4^+ (generating cubic $y_1^2 y_2 + y_2^3$, the hyperbolic umbilic) is the class whose cubic has one real linear factor, i.e. negative discriminant, and D_4^- ($y_1^2 y_2 - y_2^3$, the elliptic umbilic) the class with three; this is the convention of Proposition 6.2. The results are six.

1. **Fold.** On the open first sheet $d\,\mathrm{Exp}$ has corank one with a kernel tangent to the level sets of time (Proposition 3.2), and the fold quantity along it is one polynomial Q times an explicit nonzero factor (Theorem 4.1). A positive rational certificate for Q makes the whole open sheet a fold surface, including the tangency phases of [10, Lemma 3.3], where the time root has odd order at least three (Theorem 5.1).
2. **Edges.** The Maxwell edge is a cusp edge for every k. For $k < k_0$ the half-period edge $p = 2K$ is a second cusp edge; at $k = k_0$ its endpoint has corank two and is a hyperbolic umbilic D_4^+; for $k > k_0$ the edge $\alpha_1 = 0$, on which the two phase branches meet, is a fold (Theorem 6.1, Proposition 6.2).
3. **Atlas.** The caustic is parametrized by the midpoint amplitude B and the modulus k, with an explicit target chart (U, V, W) in which the immersion Jacobian is again Q (Proposition 7.1).
4. **Embedding.** Each of the eight open patches is embedded; the patches and their seams meet only along common source seams; as $k \downarrow 0$ the caustic collapses to the identity and as $k \uparrow 1$ it leaves every compact set (Theorems 8.3 and 8.4).
5. **Every sheet.** The kernel, the fold identity and the edge computations never use the first cell. So every later sheet is a fold surface where $Q \neq 0$, which the certificates give on every bracket; its seams are classified; the half-period seam degenerates to a hyperbolic umbilic at k_0 in every cell; and for large period count Q has an explicit leading term (Theorem 9.1, Proposition 9.2), which makes the late-fold statement of the conjugate-times paper a corollary with the sign identified.
6. **Incidences.** An octant lemma fixes which sheets can meet, the scalar level function of [11] characterizes the sheets as its integer critical levels, the incidence map of cells 1–4 is computed numerically, the modulus along the chart curve is proved unimodal, the derivative of the level function is given in closed form, and a conditional sign criterion implies the count bound at levels ≥ 1. After correction of two enclosure defects, its interval certificate was regenerated on the full $24 \times 20 \times 80$ grid: the criterion holds on

every cell outside five declared exclusion sets, the fifth being the low-modulus wedge $m \le 0.06$, $\mu \ge 0.94$, $0.41 \le \Delta\varphi \le 0.80$ together with one further cell, where the engine's budget passes leave boxes open (Section 10).

Together with the exact conjugate times this is a complete description of the first conjugate locus of SE(2) in the rotating regime, its shape, its singularities and its self-incidences, of the local structure of every later sheet, and a partial account of incidences between sheets. The count criterion is conditional and the global incidence and count claims remain open. Figure 2 shows the atlas; Figure 1 shows where the fold quantity degenerates.

The method is elementary where the original derivations used charts and minors: the first variation of the endpoint map makes the velocity transverse to the endpoint variations, so corank one reduces to the nonvanishing of one derivative of the heading, and the kernel is the null vector of the heading row. Differentiating J_2 along that kernel in raw coordinates gives the identity of Theorem 4.1, in which the transcendental Jacobi epsilon of the midpoint argument cancels. The edge classification is then a few lines each. What is not elementary is the positivity of Q: no short proof is known, and the certificate of Section 5 stands in for it.

2 Coordinates and the first sheet

We use the notation of [10]. Let $0 < k < 1$ be the modulus, $m = k^2$, $k' = \sqrt{1-m}$, $K = K(k)$ and $E = E(k)$ the complete integrals, and $\mathcal{E}(p) = \int_0^p \mathrm{dn}^2(s,k)\,ds$ Jacobi's epsilon. In the rotating coordinates

$$p = \frac{t}{2k}, \qquad \tau = \psi + p, \tag{2}$$

ψ is the initial phase and τ the midpoint phase; the terminal phase is $\psi_1 = \psi + 2p$. The source is the cylinder (ψ, k, t), $\psi \in \mathbb{R}/4K\mathbb{Z}$, for each of the two rotation signs $s_2 = \pm 1$, and the energy is $H = 1/2$. The exponential map is, for $s_2 = +1$ [12, Section 4.3],

$$\begin{aligned} x &= k\big[\mathrm{dn}\,\psi\,(\mathrm{cn}\,\psi - \mathrm{cn}\,\psi_1) + \mathrm{sn}\,\psi\,\Delta\big],\\ y &= m\,\mathrm{sn}\,\psi\,(\mathrm{cn}\,\psi - \mathrm{cn}\,\psi_1) - \mathrm{dn}\,\psi\,\Delta,\\ \theta &= \arcsin(k\,\mathrm{sn}\,\psi) - \arcsin(k\,\mathrm{sn}\,\psi_1),\\ \psi_1 &= \psi + 2p, \qquad \Delta = 2p + \mathcal{E}(\psi) - \mathcal{E}(\psi_1), \end{aligned} \tag{3}$$

and $s_2 = -1$ changes (x, y, θ) into $(-x, -y, \theta)$. The heading formula uses $(\arcsin(k\,\mathrm{sn}\,u))' = k\,\mathrm{cn}\,u$ and $\dot\theta = -\,\mathrm{cn}(\psi + t/k)$.

Sachkov's reduced Jacobian is

$$J_2 = \alpha\,\mathrm{sn}^2\,\tau + \beta\,\mathrm{cn}^2\,\tau, \qquad \alpha = (1-m)\,\mathrm{sn}\,p\,\alpha_1, \qquad \beta = f_1\beta_1, \tag{4}$$

$$\begin{aligned} \alpha_1 &= \mathrm{cn}\,p\;\mathrm{dn}\,p\,[p - 2\mathcal{E}(p)] + \mathrm{sn}\,p\{\mathrm{dn}^2\,p + \mathcal{E}(p)[p - \mathcal{E}(p)]\},\\ f_1 &= \mathrm{cn}\,p\,[\mathcal{E}(p) - p] - \mathrm{dn}\,p\;\mathrm{sn}\,p, \qquad \beta_1 = \mathrm{cn}\,p\,\mathcal{E}(p) - \mathrm{dn}\,p\;\mathrm{sn}\,p, \end{aligned} \tag{5}$$

and $\det d\,\mathrm{Exp}$ in the source coordinates (t, φ, k), $\varphi = k\psi$, equals J_2 times the nonvanishing factor $-4k/[(1-m)(1-m\,\mathrm{sn}^2\,p\,\mathrm{sn}^2\,\tau)]$. Let $p_0 = p_1^1(k) \in (K, 2K)$ be the first positive zero of f_1 and $p_1 = \min\{2K, p_\alpha\}$, p_α the first zero of α_1 after p_0; $p_1 = 2K$ for $k \le k_0$ and $p_1 = p_\alpha < 2K$ for $k > k_0$, where $k_0 = 0.9089085575\ldots$ is the modulus with $K = 2E$. On (p_0, p_1) one has $\alpha > 0 > \beta$, and with

$$r = \frac{-\beta}{\alpha - \beta} \in (0,1), \qquad J_2 = 0 \iff \mathrm{sn}^2\,\tau = r, \tag{6}$$

the first-time theorem of [10] says that r increases strictly from 0 to 1 on (p_0, p_1), and that the two phase branches $\psi_\pm = \pm F(\arcsin\sqrt{r}, k) - p \pmod{2K}$ cover the phase circle $\mathbb{R}/2K\mathbb{Z}$ exactly once. Throughout we abbreviate

$$\begin{aligned}(s, c, d, e, q) &= (\operatorname{sn} p, \operatorname{cn} p, \operatorname{dn} p, \mathcal{E}(p), p - \mathcal{E}(p)),\\ (S, C, D) &= (\operatorname{sn}\tau, \operatorname{cn}\tau, \operatorname{dn}\tau), \qquad \Delta_0 = 1 - ms^2S^2.\end{aligned} \tag{7}$$

The first sheet. The *open first sheet* is

$$\Sigma = \{(\psi, k, t) :\ p_0 < p < p_1,\ \operatorname{sn}^2\tau = r(p, k)\},$$

a smooth surface (Lemma 2.1). Since J_2 depends on ψ modulo $2K$, it has two components over each (p, k) modulo $2K$, one per phase branch, and four on the source circle $\mathbb{R}/4K\mathbb{Z}$ (the sign patches of Section 7). We use the representatives $\tau = u$ and $\tau = 2K - u$, where $u = F(\arcsin\sqrt{r}, k) \in (0, K)$: there $S = \sqrt{r} > 0$ and C has opposite signs on the two branches. The other two patches are their shifts by $2K$, which negate both S and C. Its closure adds the *Maxwell edge* $p = p_0$ ($r = 0$, $S = 0$) and the *upper edge* $p = p_1$ ($r = 1$, $C = 0$), which for $k < k_0$ is $p = 2K$ and for $k > k_0$ is $p = p_\alpha$; the two meet at k_0. The first caustic is $\Sigma_1 = \operatorname{Exp}(\overline{\Sigma})$.

Riccati variables. Following [10, Lemma 3.2] put

$$a = \frac{f_1}{q}, \qquad b = -\frac{\beta_1}{e}, \qquad z = ab = -\frac{\beta}{\ell}, \qquad \ell = eq, \tag{8}$$

so that $a, b > 0$ on (p_0, p_1), $z = ab$ increases from 0 to 1, and

$$a' = \frac{d^2 + ca + ms^2a^2}{ds}, \qquad z' = \frac{d^2b(1 - z) + s^2a(1 - m + mz)}{ds} > 0. \tag{9}$$

The collapse identities $\beta = X - \ell$, $\alpha = (1 - m)X$, $X = s\alpha_1$, give

$$r = \frac{z}{(1 - m) + mz}, \qquad 1 - r = \frac{(1 - m)(1 - z)}{(1 - m) + mz}, \qquad 1 - mr = \frac{1 - m}{(1 - m) + mz}. \tag{10}$$

One more inequality will be used repeatedly: since $f_1 = -cq - ds$,

$$\begin{aligned}a &= -c - \frac{ds}{q} < -c =: g,\\ \text{hence} \qquad d^2 + ca + ms^2a^2 &> d^2 - c^2 + ms^2a^2 = s^2(1 - m + ma^2) > 0,\end{aligned} \tag{11}$$

so $a' > 0$ on (p_0, p_1).

Lemma 2.1. *$dJ_2 \neq 0$ on $\overline{\Sigma}$ minus the point $(p, \tau, k) = (2K, K, k_0)$.*

Proof. $\partial_\tau J_2 = 2SCD(\alpha - \beta)$ is nonzero on Σ ($S, C \neq 0$, $\alpha - \beta > 0$). At $S = 0$, $\partial_p J_2 = \beta' C^2 = f_1'\beta_1 \neq 0$ since p_0 is a simple zero of f_1 and $\beta_1(p_0) < 0$. At $C = 0$, $\partial_p J_2 = \alpha'$, which at p_α is $(1 - m)s\alpha_1' \neq 0$ and at $p = 2K$ is $(1 - m)(-1)\alpha_1(2K) = 2(1 - m)(K - 2E)$, zero exactly at k_0. □

3 Corank one and the kernel

Write $\eta_\psi, \eta_k \in T_q\,\mathrm{SE}(2)$ for the endpoint variations at fixed t and $\xi = \dot g$ for the velocity, $q = \mathrm{Exp}(\psi, k, t)$, and let $\lambda(t)$ be the covector of the extremal at q.

Lemma 3.1 (First variation)**.** $\langle\lambda(t), \eta_\psi\rangle = \langle\lambda(t), \eta_k\rangle = 0$ *and* $\langle\lambda(t), \xi\rangle = 2H = 1$*. Consequently*

$$\operatorname{rank} d\,\mathrm{Exp} = \operatorname{rank}[\eta_\psi, \eta_k] + 1.$$

Proof. Let ϕ_t be the Hamiltonian flow and $\varsigma = \lambda\,dq$ the Liouville form, with the symplectic form $\omega = d\varsigma = d\lambda\wedge dq$ and X_H defined by $\iota_{X_H}\omega = -dH$, so that $\dot q = \partial_\lambda H$, $\dot\lambda = -\partial_q H$. Since H is quadratic in the fiber, $\langle\lambda, \partial_\lambda H\rangle = 2H$ and $\mathcal{L}_{X_H}\varsigma = d(2H) - dH = dH$, so $\phi_t^*\varsigma = \varsigma + t\,dH$. Restricted to the fiber $T_e^*\,\mathrm{SE}(2)$, where $\varsigma = 0$, this reads $\langle\lambda(t), d\,\mathrm{Exp}_t(\delta\lambda)\rangle = t\,dH(\delta\lambda)$ for every fiber variation $\delta\lambda$ (see also [1, Ch. 8]). Our family lies on $H = 1/2$, so the right side vanishes for $\delta\lambda = \partial_\psi, \partial_k$; and $\langle\lambda, \xi\rangle = \langle\lambda, \partial_\lambda H\rangle = 2H$. Thus $\xi \notin \ker\lambda(t) \supset \mathrm{span}(\eta_\psi, \eta_k)$. □

Proposition 3.2 (Kernel)**.** *At a conjugate point* $d\,\mathrm{Exp}$ *has corank one if and only if* $(\eta_\psi, \eta_k) \neq (0, 0)$*, and then its kernel is spanned by a vector with zero time component, namely by* $v = (F_k, -F_\psi, 0)$ *for any target coordinate* F *with* $(F_\psi, F_k) \neq (0, 0)$*. On the open first sheet*

$$\theta_\psi = k\big[\operatorname{cn}\psi - \operatorname{cn}(\psi + 2p)\big] = \frac{2k\,sSdD}{\Delta_0} \neq 0, \tag{12}$$

so $d\,\mathrm{Exp}$ *has corank one on all of* Σ*, including the tangency phases and the modulus* k_0*, with kernel* $v = (\theta_k, -\theta_\psi, 0)$*.*

Proof. If $v_\psi\eta_\psi + v_k\eta_k + v_t\xi = 0$, pairing with $\lambda(t)$ gives $v_t = 0$ by Lemma 3.1; so the kernel consists of the $(v_\psi, v_k, 0)$ with $v_\psi\eta_\psi + v_k\eta_k = 0$, it is nonzero at a conjugate point because $\det[\eta_\psi, \eta_k, \xi] = 0$ forces $\eta_\psi \parallel \eta_k$, and it is one-dimensional exactly when the pair is not zero. Any row F with nonzero gradient then gives the null vector. The heading derivative is $\theta_\psi = k[\operatorname{cn}\psi - \operatorname{cn}(\psi + 2p)]$ by (3), and the addition theorem $\operatorname{cn}(\tau \mp p) = (Cc \pm SsDd)/\Delta_0$ gives (12); on Σ, $s = \operatorname{sn} p > 0$ for $K < p < 2K$ and $|S| = \sqrt{r} > 0$. □

Remark 3.3. The kernel is tangent to the level sets of physical time. This is the general fact behind Lemma 3.1: at a conjugate time the kernel of $d\,\mathrm{Exp}$ is the kernel of $d\,\mathrm{Exp}_t$ on the energy level, since Exp is an immersion in t. It explains why the kernel in the half-angle chart is the cross product $dt \times dW$ of the time and the half-angle differentials.

4 The fold identity

Let Q be the polynomial of nineteen terms

$$\begin{aligned} Q(a, b, c, s, m) = {} & -3a^3b^2cm - a^2b^3cm + 3a^2b^2ms^2 - 8a^2b^2m + 3a^2b^2 \\ & + 2a^2bcm + a^2ms^2 - a^2s^2 + ab^3ms^2 - ab^3 - 2ab^2cm \\ & - 5abms^2 + 8abm + 3abs^2 - 6ab + acm - ac + 3bcm - 3bc, \end{aligned} \tag{13}$$

evaluated at the Riccati variables (8) and $(c, s, m) = (\operatorname{cn} p, \operatorname{sn} p, k^2)$.

Theorem 4.1 (Fold identity; computer-assisted)**.** *On the open first sheet, along the kernel* $v = (\theta_k, -\theta_\psi, 0)$ *and at fixed* t*, the following identity holds; its final reduction is a computer-algebra verification in the Jacobi ring (see the proof).*

$$v(J_2) = -\frac{2(1-m)\,q^4e^4\,S}{s\,d\,D\,\Delta_0\,(\alpha-\beta)^2}\,Q. \tag{14}$$

Apart from the displayed minus sign and the signed factor S, every factor multiplying Q is positive. Hence a point of Σ is a fold of Exp *if and only if $Q \neq 0$, and $v(\det d\,\mathrm{Exp})$ has the sign of $-SQ$ times the sign of the prefactor in* (4)*. On the chosen $S > 0$ representatives this reduces to the sign of $-Q$ times that prefactor; the phase-shift symmetries preserve the fold type.*

Proof. In the coordinates (τ, p, k) of (2), $\partial_\psi|_{k,t} = \partial_\tau$ and $\partial_k|_{\psi,t} = \partial_k|_{\tau,p} - (p/k)(\partial_\tau + \partial_p)$, so

$$v(J_2) = \theta_k\,\partial_\tau J_2 - \theta_\psi\big[\partial_k J_2 - (p/k)(\partial_\tau J_2 + \partial_p J_2)\big].$$

Here $\partial_\tau J_2 = 2SCD(\alpha - \beta)$, $\partial_p J_2$ follows from $s' = cd$, $c' = -sd$, $d' = -msc$, $e' = d^2$, $q' = ms^2$, and ∂_k at fixed argument uses the classical modulus derivatives

$$\partial_k \,\mathrm{sn} = \frac{msc^2 - \gamma cd}{k(1-m)}, \qquad \partial_k \,\mathrm{cn} = \frac{s(\gamma d - msc)}{k(1-m)},$$
$$\partial_k \,\mathrm{dn} = \frac{ks(\gamma c - sd)}{1-m}, \qquad \partial_k \mathcal{E} = \frac{k(scd - c^2 e - (1-m)ps^2)}{1-m},$$

with $\gamma = \mathcal{E}(u) - (1-m)u$ at the argument u. The heading derivative at fixed (ψ, t) is, from (3) and $(s + k\partial_k s)/d = (sd - \gamma c)/(1-m)$,

$$\theta_k = \frac{s_0 d_0 - s_1 d_1 + (\gamma_1 - \gamma_0)c_1}{1-m} - \frac{\gamma_0\,\theta_\psi}{k(1-m)} + 2pc_1, \tag{15}$$

where (s_0, c_0, d_0) and (s_1, c_1, d_1) are the Jacobi functions at $\tau - p$ and $\tau + p$ (in this proof only), given by the addition theorems, and γ_0, γ_1 the values of γ there; the addition theorem for $\mathcal{E}$ gives $\gamma_1 - \gamma_0 = 2\gamma_p - 2msS^2cd/\Delta_0$ and $\gamma_0 = \gamma_\tau - \gamma_p + mSs\,s_0$ with $\gamma_p = e - (1-m)p$. The midpoint quantity $\gamma_\tau = \mathcal{E}(\tau) - (1-m)\tau$ enters θ_k and $\partial_k \,\mathrm{sn}\,\tau$ with opposite signs and cancels identically in $v(J_2)$, which is therefore a rational function of $(s, c, d, e, q, S, C, D, m, k)$ with denominator $\Delta_0^2 k(1-m)CD$ and a numerator of 155 terms in normal form modulo $c^2 = 1 - s^2$, $d^2 = 1 - ms^2$, $C^2 = 1 - S^2$, $D^2 = 1 - mS^2$, $k^2 = m$. Substituting $S^2 = r$ from (6) leaves the single monomial SC, so that $v(J_2) = SP_\Sigma/[(\alpha - \beta)^3 \Delta_0^2 k(1-m)D]$ with P_Σ a polynomial in the p-variables, and with $\Delta_0(\alpha - \beta) = \alpha - d^2\beta$ the claim (14) becomes the polynomial identity

$$P_\Sigma\, s\, d = -2kqe(1-m)^2(\alpha - d^2\beta)\, q^3 e^3 Q,$$

which holds exactly in the ring (`research/e74_step2_fold_identity.py`; the script also checks θ_k, θ_ψ and $v(J_2)$ against finite differences of (3) to 3×10^{-20}). Positivity of the factors: $1-m, q, e, d, D, \Delta_0 > 0$, $s > 0$ on $(K, 2K)$, $|S| = \sqrt{r} > 0$, and $(\alpha - \beta)^2 > 0$. On the two chosen representatives $S > 0$; on their $2K$ shifts the identity carries the corresponding negative sign of S. Finally $\det d\,\mathrm{Exp} = FJ_2$ with $F \neq 0$ and $v(FJ_2) = Fv(J_2)$ on $J_2 = 0$. □

5 The certificate and the fold theorem

The polynomial Q is positive on the open first sheet. No short proof is known, and the positivity rests on an exact certificate; its variables are set in sans-serif to keep them apart from the coordinates.

Theorem 5.1 (Fold-only open first sheet)**.** *$Q > 0$ on the open first sheet and on its edge $\alpha_1 = 0$ for $k > k_0$. Consequently the open first sheet, including the tangency phases, consists of folds of* Exp*, and so does the edge $\alpha_1 = 0$ for $k > k_0$.*

Proof. With $g = -c$, $\rho = sd$, $\mathsf{u} = g - a = \rho/q$, $\mathsf{v} = b - g = \rho/e$, $L_0 = (1-m)/m$, the bracket inequalities give $0 < \mathsf{u} < g < 1$, $\mathsf{v} > 0$, $ab \le 1$, and

$$me - (1-m)q = m\int_0^p \mathrm{cn}^2 > 0, \qquad e > E > 1,$$

so $L_0\mathsf{v} < \mathsf{u}$ and $\mathsf{v} < \rho$. Put $\mathsf{x} = \mathsf{u}/g$, $\mathsf{t} = L_0\mathsf{v}/\mathsf{u}$, $\mathsf{y} = g^2$; then $0 < \mathsf{x}, \mathsf{t}, \mathsf{y} < 1$, and $\mathsf{v} < \rho$, $ab \le 1$ read

$$\mathsf{B} = L_0^2(1-\mathsf{y})(L_0+\mathsf{y}) - (1+L_0)\mathsf{x}^2\mathsf{t}^2\mathsf{y} > 0, \qquad \mathsf{Z} = L_0 - \mathsf{y}(1-\mathsf{x})(L_0+\mathsf{xt}) \ge 0, \tag{16}$$

with $\mathsf{Z} = 0$ exactly at $ab = 1$, i.e. at $\alpha_1 = 0$ by (8). Direct substitution gives

$$Q = \frac{g^2\mathsf{x}}{L_0^3(1+L_0)}\,P(L_0, \mathsf{x}, \mathsf{t}, \mathsf{y}),$$

and exact rational arithmetic verifies

$$P = S_0 + \mathsf{B}\,S_B + \mathsf{Z}\,S_Z, \tag{17}$$

where S_0, S_B, S_Z are sums of products $L_0^{i_0}\mathcal{B}_{i_1}^{n_1}(\mathsf{x})\mathcal{B}_{i_2}^{n_2}(\mathsf{t})\mathcal{B}_{i_3}^{n_3}(\mathsf{y})$ of Bernstein basis functions [4, 7, 5] with 216 strictly positive rational coefficients, of degree tuples at most $(5,5,4,3)$, $(2,3,2,1)$, $(4,3,3,2)$ in $(L_0, \mathsf{x}, \mathsf{t}, \mathsf{y})$. The sparse table is `data/open_sheet_q_certificate.json` (its digest is given in the code and data section); it has been re-verified independently with exact fractions only (`scripts/review/verify_q_certificate_fractions.py`). In the open cube every Bernstein function is positive, so $S_0 > 0$, and $\mathsf{B} > 0$, $\mathsf{Z} \ge 0$ give $P > 0$, including at $\mathsf{Z} = 0$. Theorem 4.1 concludes; on the edge $\alpha_1 = 0$ the point is an interior point of the smooth singular surface (Lemma 2.1) where the two phase branches meet ($C = 0$, $S = 1$), Proposition 3.2 applies, and (14) holds with $S = 1$: both sides are real-analytic on the smooth surface $\{J_2 = 0\}$ through the edge (Lemma 2.1), and the edge is nowhere dense in it, so the identity proved on the open sheet persists there. □

6 The edges

For a corank-one map with determinant $\mathcal{J}$, smooth singular set and a smooth kernel extension v near a singular point, $v(\mathcal{J}) \neq 0$ is the fold criterion, and $v(\mathcal{J}) = 0$, $v^2(\mathcal{J}) \neq 0$, $d\mathcal{J} \wedge d(v\mathcal{J}) \neq 0$ is the criterion for a cusp A_2 (Morin's criterion, [16, Appendix A, Theorem A.1]). Here A_1 (fold) and A_2 (cusp) use the Morin map convention; the corresponding generating-function types are A_2 and A_3. We use $\mathcal{J} = J_2$, which differs from $\det d\,\mathrm{Exp}$ by a nonvanishing factor, harmless for both criteria.

Theorem 6.1 (Edges). *(i) On the Maxwell edge $\{S = 0,\ f_1 = 0\}$, for every $0 < k < 1$, $\eta_\psi = 0$, the kernel is the phase direction, and the point is an A_2 cusp.*

(ii) At $(p, \tau) = (2K, K)$, $\eta_\psi = 0$, $d\theta = 0$, and

$$\eta_k = 2(e-q)\Big(1, -\frac{k}{k'}, 0\Big), \qquad e = 2E,\ q = 2(K-E).$$

Hence $d\,\mathrm{Exp}$ has corank one for $k \neq k_0$ and corank two at k_0. For $k \neq k_0$ the point is an A_2 cusp; for $k < k_0$ it is the upper edge of the first sheet.

(iii) For $k > k_0$ the edge $\alpha_1 = 0$ is a fold (Theorem 5.1).

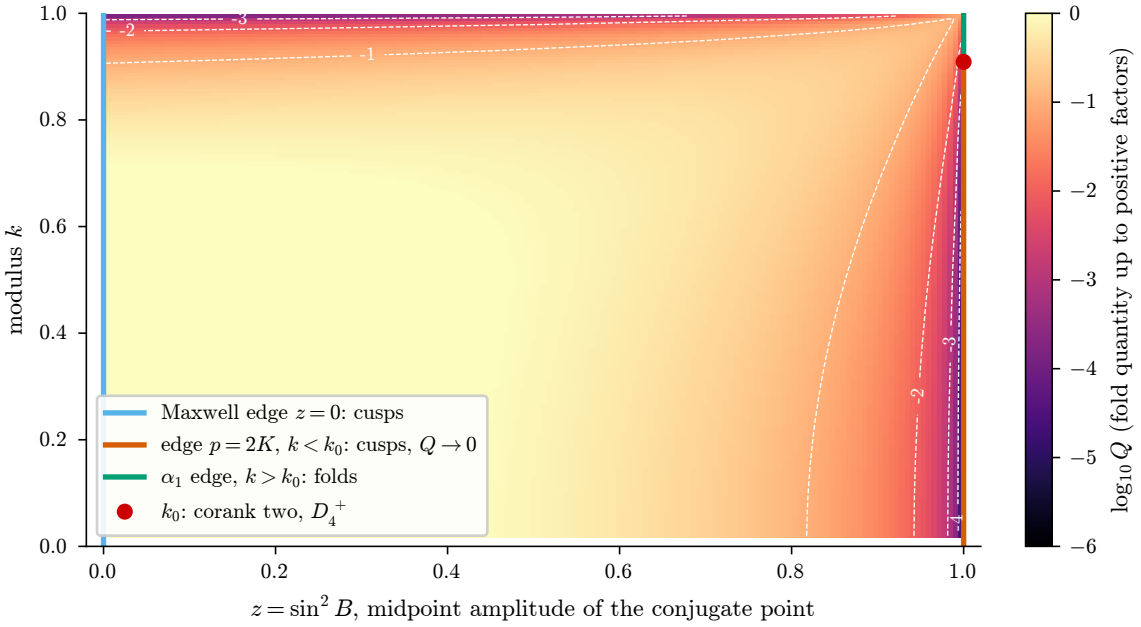


Figure 1: The fold quantity on the source rectangle of the first sheet, $z = \sin^2 B = r$ against k: $\log_{10} Q$ (color and white contour labels). Q is positive inside and tends to zero at the edge $p = 2K$ for $k \le k_0$, where the kernel derivative of J_2 must vanish because that edge is a cusp edge, and along the separatrix $k \to 1$. At the Maxwell edge $z = 0$ it is the factor $S = \sqrt{z}$ in (14), not Q, that vanishes. The α_1 edge above k_0 carries $Q > 0$: it is a fold.

Proof. (i) At $S = 0$ one has $\Delta_0 = 1$, $D = 1$, $C = \pm 1$, and from (3) by the addition theorems

$$x_\psi = -2kC\,d\,f_1, \qquad y_\psi = 2ms\,f_1, \qquad \theta_\psi = 0, \qquad \theta_k = \frac{2C\beta_1}{1-m};$$

so on the edge $\eta_\psi = 0$, and $\theta_k \neq 0$ because $\beta_1 < 0$ on (K, b_1): corank one, kernel ∂_ψ, by Proposition 3.2. The extension $v = (\theta_k, -\theta_\psi, 0)$ is smooth through the edge and equals the kernel there. On the sheet $v(J_2) = SG$ by (14), with $G \neq 0$ at the edge: $f_1 = 0$ gives $a = 0$ and $Q = -3(1-m)bc > 0$. So $v(J_2) = 0$ at the edge, and v is tangent to the sheet there. A derivative along a vector tangent to the sheet depends only on the restriction to the sheet, hence

$$v^2(J_2) = dS(v)\,G = \frac{2\beta_1}{1-m}\,G = -\frac{12(1-m)q^4e^3\beta_1^2\,c}{s\,d\,\alpha^2} > 0,$$

using $dS(v) = \theta_k CD = 2\beta_1/(1-m)$ and, at the edge, $\beta = 0$, $\Delta_0 = D = 1$, $Q = 3(1-m)c\beta_1/e$. The same tangent derivative shows $d(vJ_2)(v) \neq 0$ while $dJ_2(v) = 0$, which is the rank condition.

(ii) At $(2K, K)$: $s = 0$, $c = -1$, $d = 1$, $C = 0$, $S = 1$, $D = k'$, $\theta = 0$. From (3), $x = -2kqS$ and $y = -2qD$ along $s = 0$, so $x_\psi = -2kqCD = 0$ and $y_\psi = 2mqSC = 0$; $\theta_\psi = 0$ by (12) and $\theta_k = 0$ by (15); $\dot\theta = -\operatorname{cn}(3K) = 0$. The modulus derivatives at fixed (ψ, t) are $x_k = 2(e-q)$ and $y_k = 2k(q-e)/k'$ (`research/e74_step3_edges.py`, exact). Since $e - q = 2(2E - K)$ vanishes exactly at k_0, Lemma 3.1 gives the corank statement. For $k \neq k_0$ take the extension $v = (x_k, -x_\psi, 0)$ from the x row. At the point $v(J_2) = x_k\partial_\tau J_2 = 0$ because $C = 0$, and v is tangent to the sheet ($\partial_\tau J_2 = 0$, $\partial_p J_2 \neq 0$). Differentiating once more along $v = x_k\partial_\tau$ at the point,

$$v^2(J_2) = x_k\left[x_k\,\partial_\tau^2 J_2 - x_{\psi\psi}\,\partial_k^{\mathrm{src}} J_2\right] = -12eq(1-m)(e-q)^2 \neq 0,$$

from $\partial_\tau^2 J_2 = -2eq(1-m)$, $x_{\psi\psi} = 2kq(1-m)$ and $\partial_k^{\mathrm{src}} J_2 = (e-q)e/k$, where $\partial_k^{\mathrm{src}}$ is the k-derivative at fixed (ψ, t). The bracket, $-6eq(1-m)(e-q)$, is $d(vJ_2)(\partial_\tau)$ while $dJ_2(\partial_\tau) = 0$: the rank condition. For $k < k_0$ the point is the upper end $p = p_1 = 2K$ of the first sheet; for $k > k_0$ it lies on the closure of the right sheet of cell 1, Σ_1^+ in the notation of Section 9. □

Proposition 6.2 (Hyperbolic umbilic; computer-assisted)**.** *At $k = k_0$ the point $(p, \tau) = (2K, K)$ is a D_4^+ Lagrangian singularity of* Exp.

Proof. Exp is Lagrangian: homogeneity identifies (λ, t) on the energy cylinder with $t\lambda$ in the fiber, the flow maps the fiber to a Lagrangian submanifold, and Exp is its projection. For the germ we use the amplitude chart $(A, B, k) = (\operatorname{am} p, \operatorname{am} \tau, k)$ and the half-angle target chart (U, V, W) of Proposition 7.1 below. Here $e = \mathcal{E}(2K) = K(k_0) > 1$ and $k' = \sqrt{1-k_0^2}$. The vectors $(1/k, 0, 0)$, $(0, 1, 0)$, $(-e, 0, k(1-k^2))$ have $dV = 1$ on the first while the last two span the kernel, and the image of $d\,\mathrm{Exp}$ is the V-axis, so (U, W) are cokernel coordinates. Hessian contractions of the chart formulas (20), computed by computer algebra and confirmed numerically to 10^{-10}, after straightening the regular coordinate $\sigma = V - V_0$ and removing pure σ^2 target terms, give the singular quadratic components in kernel coordinates (y_1, y_2)

$$U_2 = \mathsf{a} y_1^2 + \mathsf{b} y_2^2 + \mathsf{c}\,\sigma y_2, \qquad W_2 = \mathsf{d} y_1 y_2 + \mathsf{f}\,\sigma y_1,$$

$$\mathsf{a} = \frac{ek^2}{2k'}, \quad \mathsf{b} = \frac{ek^2 k'(e^2-1)}{2}, \quad \mathsf{c} = -e^2 k k', \quad \mathsf{d} = ek, \quad \mathsf{f} = -1.$$

Since Exp is Lagrangian, the intrinsic second derivative $\ker \to \operatorname{coker}$ is the gradient of a binary cubic under the symplectic pairing of the kernel with the cokernel; let (m_{ij}) be that pairing, which turns (U_2, W_2) into a gradient in (y_1, y_2) (it is unique up to scale, and the sign of the discriminant is scale-invariant); Hessian symmetry forces $m_{12}\mathsf{d} = 2m_{21}\mathsf{a}$, $m_{22}\mathsf{d} = 2m_{11}\mathsf{b}$, $m_{11}\mathsf{c} = m_{22}\mathsf{f}$, and $\mathsf{c} - 2\mathsf{bf}/\mathsf{d} = -kk' \neq 0$ gives $m_{11} = m_{22} = 0$. With $m_{21} = 1$ the generating-family jet is

$$\widehat{F}_3 = \mathsf{a} y_1^2 y_2 + \frac{\mathsf{b}}{3} y_2^3 + \sigma\Big(\frac{\mathsf{af}}{\mathsf{d}} y_1^2 + \frac{\mathsf{c}}{2} y_2^2\Big),$$

whose binary cubic at $\sigma = 0$ has discriminant $-4\mathsf{a}^3\mathsf{b}/3 = -e^4k^8(e^2-1)/(12k'^2) < 0$: the D_4^+ sign [2, Ch. 5, §2.3, p. 97]. The Jacobi algebra is spanned by $1, y_1, y_2, y_1^2$; the two cokernel coordinates represent y_1, y_2 and the regular parameter represents y_1^2 with coefficient $\mathsf{a}(\mathsf{f}/\mathsf{d} - \mathsf{c}/(2\mathsf{b})) = k/[2k'(e^2-1)] > 0$, so the three-parameter deformation is versal [2, Ch. 5, §2.5, pp. 98–99]. For completeness, the cubic is 3-right-determined. After positive linear rescaling it is $f = x^2 y + y^3$. The five polynomials $x^2 f_x, xyf_x, y^2 f_x, x^2 f_y, y^2 f_y$ form a basis of the quartic homogeneous polynomials. If R vanishes to order four, the same five expressions for $f_t = f + tR$ have an invertible coefficient matrix over the quartic monomials near the origin, uniformly for $0 \le t \le 1$. Thus $R = df_t(X_t)$ for a smooth vector field X_t vanishing to order two. The local flow of $-X_t$ fixes the origin and satisfies $f_t \circ \Phi_t = f$, proving the assertion. Together with versality, this identifies the full germ, by the generating-family equivalence theorem [2, Ch. 5, §1.4, p. 93]. The same three local types occur on the ellipsoid [8]. □

Remark 6.3 (Intrinsic check). The real form of the generating cubic can be checked intrinsically; classification of the full germ still requires the versality established above. On the kernel plane $\ker d\,\mathrm{Exp} = \operatorname{span}(\partial_\psi, \partial_k)$ the second differential modulo the image $\mathbb{R}\xi$ is a quadratic map $\Phi : \ker \to \operatorname{coker}$, which for a Lagrangian map is the gradient of the generating cubic in suitable linear coordinates; the quadratic form $v \mapsto \det D\Phi(v)$ is then the Hessian form of that cubic up to a constant, definite for D_4^- and indefinite for D_4^+. At

40 digits its discriminant at our point is $2.2 \times 10^5 > 0$ (`research/e74_step4_d4.py`), in agreement with the proposition. This discriminant alone is insufficient: the Lagrangian maps $(\sigma, 2xy, x^2+3y^2)$ and $(\sigma, 2xy+2\sigma x, x^2+3y^2)$ have the same intrinsic quadratic map at the origin, but their rank-one loci are respectively a line and an isolated point. Only the second has the missing quadratic parameter class. The local-algebra and rank-locus checks are exact (`research/e80_singularity_recognition.py`).

7 The atlas

Let $B = \operatorname{am}(\tau, k)$ be a continuous amplitude and $z = \sin^2 B$; by the first-time theorem $r(\cdot, k)$ has a unique inverse $\mathcal{P}(z, k)$ on (p_0, p_1), and

$$\tau = F(B,k), \qquad \psi = \tau - \mathcal{P}(\sin^2 B, k), \qquad t = 2k\mathcal{P}(\sin^2 B, k) \tag{18}$$

parametrizes the first singular surface, one full period of B covering the four sign patches for one sign s_2. The half-angle rectifying chart of [12] is

$$W = \tan\frac{\theta}{2}, \qquad U = -\frac{y\cos\frac{\theta}{2} - x\sin\frac{\theta}{2}}{2\cos\frac{\theta}{2}}, \qquad V = -\frac{x\cos\frac{\theta}{2} + y\sin\frac{\theta}{2}}{2\cos\frac{\theta}{2}}, \tag{19}$$

a diffeomorphism of $\{|\theta| < \pi\}$ onto $\mathbb{R}^3$.

Proposition 7.1 (Stratified atlas)**.** *With $q = \mathcal{P} - \mathcal{E}(\mathcal{P})$, $s = \operatorname{sn}\mathcal{P}$, $d = \operatorname{dn}\mathcal{P}$, $D = (1 - k^2\sin^2 B)^{1/2}$, the caustic for $s_2 = +1$ is*

$$U = \frac{qD}{d}, \qquad V = \frac{k f_1(\mathcal{P})\sin B}{d}, \qquad W = -\frac{ks\cos B}{d}. \tag{20}$$

The other patches are $B \mapsto -B : (U,V,W) \mapsto (U,-V,W)$, $B \mapsto \pi - B : (U,V,-W)$, $B \mapsto B + \pi : (U,-V,-W)$, and $s_2 \mapsto -s_2 : (-U,-V,W)$. On every open patch (U,V) are local coordinates, and on $0 < B < \pi/2$

$$\det\frac{\partial(U,V)}{\partial(z,k)} < 0. \tag{21}$$

Proof. (20) is (3) in the chart (19), as an identity in (p, τ, k) with $p = \mathcal{P}$, $\sin B = \operatorname{sn}\tau$: with $\psi = \tau - p$, $\psi_1 = \tau + p$, $S = \operatorname{sn}\tau$ and $\Delta_0 = 1 - mS^2s^2$, the addition theorems give $\operatorname{cn}\psi - \operatorname{cn}\psi_1 = 2SsDd/\Delta_0$, $\operatorname{sn}\psi = (Scd - CDs)/\Delta_0$, $\operatorname{dn}\psi = (Dd + mSsCc)/\Delta_0$, and the addition theorem for $\mathcal{E}$ gives $\Delta = 2q + 2mS^2scd/\Delta_0$; with $\sin\theta = k(\operatorname{sn}\psi\operatorname{dn}\psi_1 - \operatorname{dn}\psi\operatorname{sn}\psi_1)$, $\cos\theta = \operatorname{dn}\psi\operatorname{dn}\psi_1 + m\operatorname{sn}\psi\operatorname{sn}\psi_1$ and $W = \sin\theta/(1+\cos\theta)$, substitution into (19) reduces, modulo $c^2 + s^2 = 1$, $d^2 = 1 - ms^2$ and the same relations at τ, to (20) (`research/e79_atlas_identity.py`, symbolic; the formulas hold off the sheet as well). The sign actions are Jacobi parity and periodicity. The restriction of $d\operatorname{Exp}$ to the tangent plane of Σ is injective exactly when the kernel v is transverse to Σ, i.e. when $v(J_2) \neq 0$: that is the fold property, Theorem 5.1, so the caustic is an immersed surface. Its projection determinant $\partial(U,V)/\partial(z,k)$ reduces on $J_2 = 0$, after factors of fixed sign, to the same polynomial P as the certificate (Proposition A.1 in the appendix), hence is nonzero by Theorem 5.1; so (U,V) are local coordinates on each open patch, and the sign in (21) is fixed by continuity and one evaluation (at $k = 0.8$, $B = \pi/4$). □

8 Embedding and incidence

Lemma 8.1 (Slope)**.** *On the fundamental patch, write $\zeta = ab$ for the Riccati variable of (8), retaining $z = \sin^2 B = r$ for the atlas coordinate. Then $H = V/U$ satisfies*

$$H = \frac{k}{k'}\, a\sqrt{\zeta}, \qquad \frac{\partial \log H}{\partial p} = \frac{a'}{a} + \frac{\zeta'}{2\zeta} > 0. \tag{22}$$

Since $z = r = \zeta/(1 - m + m\zeta)$ increases with ζ, also $H_z > 0$. At fixed k the slope runs once from 0 at the Maxwell seam to an upper value $H_1(k)$ at the upper seam.

Proof. $V/U = kf_1 S/(qD)$ and by (10) $S^2/D^2 = r/(1 - mr) = \zeta/(1 - m)$. The derivative is positive by (9) and (11); and p increases with the atlas coordinate $z = r$. □

Lemma 8.2 (Upper slope)**.** *$H_1(k) = k/k'$ for $k \le k_0$, and H_1 is strictly increasing on $(0, 1)$, with $H_1 \to 0$ as $k \downarrow 0$ and $H_1 \to \infty$ as $k \uparrow 1$ (for $k > k_0$ the sign of H_1' rests on an exact differentiation replayed in `scripts/first_caustic_global_univalence_proof.py`).*

Proof. For $k \le k_0$ the upper seam is $p = 2K$, where $f_1 = q = 2(K - E)$ and $z = 1$. For $k > k_0$ it is $\alpha_1 = 0$, where $z = 1$ and $H_1 = (k/k')\, f_1/q$. The argument is as follows. Set $\varpi = 2K - p_\alpha$ and, at ϖ, $\tilde{s} = \operatorname{sn}\varpi$, $\tilde{g} = \operatorname{cn}\varpi$, $\tilde{d} = \operatorname{dn}\varpi$, $\tilde{e} = 2E - \mathcal{E}(\varpi)$, so that $e = \tilde{e}$ and $q = p_\alpha - \tilde{e}$ at p_α; the endpoint equation $\alpha_1(p_\alpha) = 0$ reads $q(\tilde{e}\tilde{s} - \tilde{g}\tilde{d}) + \tilde{d}(\tilde{g}\tilde{e} + \tilde{s}\tilde{d}) = 0$. With $\upsilon = \tilde{d}/\tilde{e}$ and $\varrho = \tilde{e}\tilde{s}/(\tilde{d}\tilde{g}) \in (0, 1)$,

$$\frac{f_1}{q} = \frac{\sqrt{1 + \varrho^2\upsilon^2}}{1 + \varrho\upsilon^2}, \qquad \frac{q}{\tilde{e}} = \frac{1 + \varrho\upsilon^2}{1 - \varrho}, \qquad \frac{p_\alpha}{\tilde{e}} = \frac{2 - \varrho(1 - \upsilon^2)}{1 - \varrho},$$

and exact differentiation along $\alpha_1 = 0$ shows that $H_1'(k)$ has the sign of $-F_- G$ with

$$-F_- = 1 + \varrho\upsilon^2(1 - \varrho) + \varrho^3\upsilon^4 + k^2\varrho^3\upsilon^2(1 - \upsilon^2) + 2(1 - k^2)\varrho^2\upsilon^2 > 0,$$

$$G = (1 - \varrho)(1 + \varrho\upsilon^2)\{1 + (2\varrho - 1)\upsilon^2\} - (1 - m)\varrho\upsilon^2\{2 - \varrho(1 - \upsilon^2)\}^2.$$

$G > 0$ follows from $G > (1 - \varrho)^2\mathcal{G}/\tilde{e}^4$ with $\mathcal{G} = \tilde{e}q(\tilde{e}^2 - \tilde{d}^2) - (1 - m)\tilde{d}^2 p_\alpha^2$, and from the three bounds $k_0^2 > 3/4$, $\tilde{e} > 3/2$, $2(1 - m)(K - E) < 3/4$ of Lemma A.2 in the appendix, which, with $\tilde{d} \le 1$, $\tilde{e} < q < 2(K - E)$ (so that $p_\alpha^2/q = q + 2\tilde{e} + \tilde{e}^2/q < 2(K - E) + 3\tilde{e}$) and $1 - m < 1/4$, give $\mathcal{G} > q\big[\tilde{e}(\tilde{e}^2 - 1) - \frac{3}{4} - \frac{3}{4}\tilde{e}\big] = \frac{q}{4}(\tilde{e} + 1)(2\tilde{e} - 3)(2\tilde{e} + 1) > 0$. The two formulas for H_1 agree at k_0; the last factor of f_1/q exceeds $1/2$, so $H_1 \to \infty$; and $H_1 \to 0$ with k. □

Theorem 8.3 (Each open patch is embedded)**.** *On every open patch the projection $(z, k) \mapsto (U, V)$ is injective; hence (U, V, W) embeds the patch.*

Proof. Take the fundamental patch. For $h > 0$ the level $H = h$ exists exactly for $\kappa(h) < k < 1$, $\kappa = H_1^{-1}$, by Lemma 8.2 and is a single graph $z = z_h(k)$ by Lemma 8.1. The quotient rule and (21) give $\det \partial(U, H)/\partial(z, k) = U^{-1} \det \partial(U, V)/\partial(z, k) < 0$, so along the level

$$\frac{dU}{dk}\Big|_{H=h} = -\frac{\det \partial(U, H)/\partial(z, k)}{H_z} > 0.$$

Two source points with the same (U, V) have the same H and U, hence the same k, hence the same z. □

Theorem 8.4 (Closed incidence)**.** *For $0 < k < 1$, the caustic is a topological embedding of every fundamental strip $0 \le B \le \pi/2$, including its seams, into* SE(2)*. This embedding is proper into the punctured target* $\mathrm{SE}(2) \setminus \{e_{\mathrm{SE}(2)}\}$*, where $e_{\mathrm{SE}(2)}$ is the identity, but not into* SE(2) *itself. The Maxwell seam is cuspidal; the upper seam is cuspidal for $k < k_0$, an umbilic at k_0, and a fold for $k > k_0$ (Theorem 6.1, Proposition 6.2). The eight open patches and their seams have no target incidences except the identification of a common source seam. As $k \downarrow 0$ the caustic collapses to the identity; no sequence with $k \uparrow 1$ stays in a compact set.*

Proof. On the lower seam $B = 0$ one has $f_1 = 0$, so $U = q/d = -s/c$, $V = 0$, $-W = ks/d = kU/\sqrt{1 + (1-m)U^2}$, and along $f_1 = 0$

$$U_k(f_1)_A - U_A(f_1)_k = \frac{k\beta_1}{c^2(k^2-1)} > 0, \qquad (f_1)_A = -\frac{d}{c} > 0,$$

($A = \operatorname{am} p$), so $dU/dk > 0$; $U \to 0$ as $k \downarrow 0$ and $U \ge q > K - 2E \to \infty$ as $k \uparrow 1$: the seam is embedded. The upper seam is embedded because its slope H_1 is strictly increasing. The signs on the three strata are $(U, V, W) = (+, +, -)$ inside, $(+, 0, -)$ on $B = 0$, $(+, +, 0)$ on $B = \pi/2$; the eight open patches occupy the eight open octants, and adjacent patches share seams: the four lower seams carry the labels $(\pm, 0, \pm)$ and the four upper seams $(\pm, \pm, 0)$, one seam per label. As $k \downarrow 0$, uniformly on the strip, $p = \pi + O(k^2)$, $q = O(k^2)$, $\operatorname{sn} p = O(k^2)$, so $(U, V, W) = O(k^2, k^3, k^3)$ and $t = 2kp \to 0$. For $k \uparrow 1$ let $R^2 = x^2 + y^2 = 4(U^2 + V^2)/(1 + W^2)$ be the spatial radius; from (20) and (10), with the Riccati variable $\zeta = ab$, $\chi = k^2 g^2 \zeta/(1-m)$ and $\varrho_0 = a/g$,

$$\frac{R^2}{4q^2} = 1 - \frac{(1-\varrho_0^2)\chi}{1+\chi} \ge 1 - \frac{2}{q},$$

because $1 - \varrho_0 = sd/(qg) \le q^{-1}\sqrt{1 + 1/\chi}$, i.e. $s^2 d^2 \le g^2 + (1-m)/(m\zeta)$, which follows from $s^2 d^2 = (1-y)(1-m+my) \le 1 - m + my \le y + (1-m)/m \le y + (1-m)/(m\zeta)$ with $y = g^2$, and $\sqrt{\chi(1+\chi)} \le 1 + \chi$; and $q \ge q(p_0) > K - 2E \to \infty$. The two boundary limits confine the inverse image of any compact subset of $\mathrm{SE}(2) \setminus \{e_{\mathrm{SE}(2)}\}$ to $[0, \pi/2] \times [\epsilon, 1-\epsilon]$ for some $\epsilon > 0$. It is closed there, hence compact. Properness and the proved injectivity give a topological embedding into the punctured target, and therefore into SE(2). Properness into SE(2) fails: at fixed B, a sequence $k_j \downarrow 0$ has images converging to $e_{\mathrm{SE}(2)}$, so these images together with $e_{\mathrm{SE}(2)}$ form a compact set whose inverse image contains a sequence without a convergent subsequence in the strip. □

9 Every sheet

By [10, Theorem 4.1], in the cell $[p_n^1, p_{n+1}^1)$, $n \ge 1$, every rotating geodesic has one conjugate time on each of the brackets (two distinct times except at $k = k_0$, $\psi \equiv K \pmod{2K}$, where both equal $2nK$) $B_n^- = [p_n^1, \min(2nK, p_n^\alpha)]$ and $B_n^+ = [\max(2nK, p_n^\alpha), b_n]$, where p_n^α is the zero of α_1 in the cell and b_n the zero of β_1 in $(2nK, (2n+1)K)$; on each open bracket $0 < r < 1$ and the conjugate points are $\operatorname{sn}^2 \tau = r$. The *sheets* $\Sigma_n^\pm$ are these two surfaces in the source, each with two mirror branches $\tau = \pm u$; their closures add the seams $S = 0$ (at p_n^1 for Σ_n^-, at b_n for Σ_n^+) and $C = 0$ (at $2nK$ or p_n^α). The first sheet is Σ_1^-. With the sign $\sigma_n = (-1)^{n+1}$ and $g = |\operatorname{cn} p|$, $\mathsf{u} = |\operatorname{sn} p| \operatorname{dn} p/q$, $\mathsf{v} = |\operatorname{sn} p| \operatorname{dn} p/e$,

$$(\sigma_n a, \sigma_n b) = (g - \mathsf{u},\ g + \mathsf{v}) \text{ on } \Sigma_n^-, \qquad (\sigma_n a, \sigma_n b) = (g + \mathsf{u},\ g - \mathsf{v}) \text{ on } \Sigma_n^+, \qquad \sigma_n c = -g, \tag{23}$$

because f_1 has the sign σ_n on the cell, β_1 the sign $(-1)^n = -\sigma_n$ on (b_{n-1}, b_n), and $\operatorname{sn} p$ the sign σ_n on the left half and $-\sigma_n$ on the right half of the cell. Figure 3 shows the first three cells.

Theorem 9.1 (Every sheet)**.** *Let $n \geq 1$ and $0 < k < 1$ (at $k = k_0$ the two brackets touch at $2nK$ and $0 < r < 1$ still holds on both open brackets, [10, Theorem 4.1]).*

(i) On the open sheets $\Sigma_n^{\pm}$ the map Exp *has corank one, its kernel is $(\theta_k, -\theta_\psi, 0)$, and the identity* (14) *holds with $S = \operatorname{sn}\tau$ and $s = \operatorname{sn} p$ signed.*

(ii) (computer-assisted) $Q > 0$ on Σ_n^- and $Q < 0$ on Σ_n^+, up to and including their α_1 ends when $k \neq k_0$. Hence every open sheet is a fold surface, and every noncritical α_1 seam is a fold. At $k = k_0$ the common end has $Q = 0$ and is covered by (iii).

(iii) The seam $S = 0$ of Σ_n^- (at p_n^1) is an A_2 cusp edge; the seam $S = 0$ of Σ_n^+ (at b_n) is a fold; the seam $C = 0$ at $p = 2nK$ is an A_2 cusp edge for $k \neq k_0$; at $(p, \tau, k) = (2nK, K, k_0)$ the map has corank two and is a hyperbolic umbilic D_4^+.

Proof. (i) Proposition 3.2 and Theorem 4.1 used only $J_2 = 0$, $\operatorname{sn} p \neq 0$ and $\operatorname{sn}\tau \neq 0$, which hold on every open sheet ($p \notin K\mathbb{Z}$ there, and $0 < r < 1$).

(ii) In the variables (23) the certificate of Theorem 5.1 applies to Σ_n^- verbatim: $0 < \mathsf{u} < g < 1$ and $\mathsf{v} > 0$ by (23); $ab \leq 1$ is $\mathsf{Z} \geq 0$ (equality at the α_1 end); $e = \mathcal{E}(p) > E > 1$ because $p > K$; and $me - (1-m)q = m\int_0^p \operatorname{cn}^2 > 0$. So $P = S_0 + \mathsf{B}S_B + \mathsf{Z}S_Z > 0$ and $Q > 0$, with $\mathsf{Z} = 0$ allowed at the end. On Σ_n^+ the orientation of (23) is reversed, and $Q < 0$ there, up to and including the α_1 end when $k < k_0$, by Proposition A.3 in the appendix: the bound $\mathsf{t} = L_0\mathsf{v}/\mathsf{u} < 1/2$ on every later cell, the concavity of the scaled $-Q$ in g^2, an elementary argument on the face $ab = 1$ and four exact rational certificates on the face $\mathsf{v} = \rho/2$ (`data/universal_right_sv_certificates.json`, replayed by `scripts/universal_right_fold_proof.py`). The hypotheses of both certificates were also checked on 4,480 interior points of the brackets of cells 1–8 at seven moduli. The minima of the five recorded diagnostic margins lie between 5×10^{-4} and 2×10^{-2} (`research/e75_step3_certificate_domains.py`); this range does not bound every polynomial margin. All points were independently replayed at 60 digits (`research/e80_certificate_domain_audit.py`). This finite survey does not test the critical endpoint. Away from $k = k_0$, the α_1 seams are interior points of the smooth singular surface where the two mirror branches meet ($C = 0$, $S = 1$), so (i) applies and they are folds by (14). At the critical end, $s = 0$, $e = q$, $a = b = -c = \pm 1$ instead give $Q = 0$.

(iii) At $S = 0$ the formulas of Theorem 6.1(i) hold for every n: $x_\psi = -2kCdf_1$, $y_\psi = 2mscf_1$, $\theta_\psi = 0$, $\theta_k = 2C\beta_1/(1-m)$. At p_n^1 therefore $\eta_\psi = 0$, the kernel is the phase direction, $Q = -3(1-m)bc \neq 0$, and $v^2(J_2) = -12(1-m)q^4e^3\beta_1^2\, c/(s\,d\,\alpha^2)$, nonzero because $c/s < 0$ on $((2n-1)K, 2nK)$: a cusp, with the rank condition as before. At b_n one has $\theta_\psi = \theta_k = 0$ but $x_\psi \neq 0$, so the x-row extension $v = (x_k, -x_\psi, 0)$ is the kernel, and with $\beta_1(b_n(k), k) = 0$ an exact computation in the Jacobi algebra (`research/e74_step3_edges.py`, extended to b_n by quasi-periodicity) gives

$$v(J_2) = 2kC^3d\,f_1^2\,\partial_k^{\mathrm{src}}\beta_1 = \frac{2k^2C^3d\,f_1^2}{mc}\,(es^2 - qc^2), \tag{24}$$

where $\partial_k^{\mathrm{src}}$ is the derivative at fixed (ψ, t). With the signs normalized so that $s, c, d > 0$, the seam equation $ce = ds$ gives $c^2 = d^2/(e^2 + d^2)$ and $d^2(d^2 + e^2 - 1) = (1-m)e^2$, hence $es^2 - qc^2 = (e^3 - qd^2)/(e^2 + d^2)$, and $qd^2 < e^3$ follows from $(1-m)q < (1-m)(2n+1)K \leq (2n+1)\pi/2 < 2n(4n^2-1) < e(e^2-1)$, using $e > 2nE > 2n$: the seam is a fold. At $(2nK, K)$, $\eta_\psi = 0$, $d\theta = 0$ and $\eta_k = 2(e-q)\big((-1)^{n+1}, -k/k', 0\big)$ with $e = 2nE$, $q = 2(K-E)n$, so $e - q = 2n(2E - K)$ vanishes exactly at k_0; for $k \neq k_0$ the computation of Theorem 6.1(ii) goes through with the sign $(-1)^{n+1}$ cancelling in $v^2(J_2) = -12eq(1-m)(e-q)^2$: a cusp. At k_0 the generating family of Proposition 6.2 applies with $e = \mathcal{E}(2nK) = nK > 1$ (for even n after the target reflection $(V, W) \mapsto (-V, -W)$), its discriminant $-e^4k^8(e^2-1)/(12k'^2)$ and versal coefficient $k/[2k'(e^2-1)]$ keeping their signs; the intrinsic test of Remark 6.3

confirms D_4^+ at $n = 1, 2, 3$; the closed forms hold for every n because at $p = 2nK$, $\tau = K$ the Jacobi values are those at $p = 0$ up to the sign $(-1)^n$ (absorbed by the reflection for even n), and $q = e = nK$ at k_0, so the contractions depend on (e, k) only. □

Proposition 9.2 (Leading term of Q)**.** *Put $g = -\sigma_n c$ and $\mathsf{u} = g - \sigma_n a$, $\mathsf{v} = \sigma_n b - g$, so that $\mathsf{u} = \rho/q$, $\mathsf{v} = \rho/e$ with $\rho = \sigma_n \operatorname{sn} p \operatorname{dn} p$. Then Q is a polynomial in (u, v) of degree at most three in each variable, with twelve terms and coefficients in (g, m), vanishing at $\mathsf{u} = \mathsf{v} = 0$, with linear part $4g(1 - g^2)(1 - m + mg^2)\,\mathsf{u}$ and no linear term in v. On the cell n at $p = 2nK + \sigma$, $|\sigma| < K$, exactly $q = 2n(K - E) + \sigma - \mathcal{E}(\sigma)$ and $e = 2nE + \mathcal{E}(\sigma)$, hence*

$$Q = \frac{2 \operatorname{sn}^3 p \operatorname{dn} p\,(-\operatorname{cn} p)\,(1 - m + m \operatorname{cn}^2 p)}{n\,(K - E)} + O(n^{-2}), \tag{25}$$

uniformly for σ in compact subsets of $(-K, 0) \cup (0, K)$. Its sign is that of $-\operatorname{sn}\sigma \operatorname{cn}\sigma$: positive on the left half of the cell, negative on the right half, at every modulus.

Proof. Substituting $a = \sigma_n(g - \mathsf{u})$, $b = \sigma_n(g + \mathsf{v})$, $c = -\sigma_n g$ into (13) and reducing with $s^2 = 1 - g^2$ gives the twelve-term polynomial; its constant term vanishes and its linear part is as stated (`research/e80_late_polynomial.py`, exact). Quasi-periodicity gives the exact q and e, so $\mathsf{u}, \mathsf{v} = O(1/n)$ and the linear term dominates. □

Corollary 9.3 (Late conjugate points are folds)**.** *The conjugate points of every cell $n \geq 1$ are folds, for every $0 < k < 1$ and every phase off the seams. On every compact set of (ψ, k) with $\psi \not\equiv K$ there is n_0 such that for $n \geq n_0$ this follows from* (25) *alone, without the certificates.*

Proof. The first sentence is Theorem 9.1(ii). By the large-index law [10, Theorem 5.1(ii)] the offsets σ of the two conjugate points of cell n converge to $\mp K/2 - \psi/2$, with ψ taken in $(-K, K)$, uniformly on the set (the remainders of the expansions behind that law are explicit continuous functions of (σ, k)), so for $n \geq n_0$ they lie in a compact subset of $(-K, 0) \cup (0, K)$, where (25) has the sign of $-\operatorname{sn}\sigma \operatorname{cn}\sigma$ and its remainder is smaller. □

Remark 9.4. For $k \neq k_0$ the sign of Q changes between the two brackets inside the gap between $2nK$ and p_n^α, in either order. Put $D_0 = K - E$ and $\sigma = w/n$. The leading polynomial balance is

$$n^4 Q \approx w^2 \left[-\frac{2w}{D_0} + \frac{3(2E - K)}{4D_0^2 E} \right], \qquad n\sigma_\alpha \approx \frac{2E - K}{2D_0 E}.$$

Its nonzero root is three quarters of the endpoint offset. The numerically computed argument-offset ratios $(p_Q - 2nK)/(p_n^\alpha - 2nK)$ are 0.771, 0.755, 0.7503 at $n = 2, 5, 20$ for $k = 0.6$; these are not ratios of Jacobi sn values. The balance is checked in `research/e80_late_polynomial.py`; we do not use it as a certificate in the shrinking gap, since the fold certificates already cover every open sheet. The two cusp seams lie on the closures of the reflection Maxwell strata MAX_2 ($p = p_j^1$) and MAX_6 ($p = 2jK$) of [12, Theorem 6.2], the fold seams on neither.

10 Incidences between sheets

Two sheets meet where a point of SE(2) is a conjugate point along two different geodesics. The chart (19) first restricts which sheets can meet at all, and the scalar inverse of the endpoint map [11] then reduces the question to the level function of a target.

Notation. In this section (A_q, B_q) are the coordinates of the target in the frame of the heading bisector (the amplitudes $A = \operatorname{am} p$, $B = \operatorname{am} \tau$ of Sections 7–8 do not occur here), ν is the parameter of the chart curve (the kernel v of Section 3 does not occur), $L(m) = 4(K(m) - E(m))$ is the drift per period of [11], written L when the argument is clear, and $\beta_N = 4E(m)/L(m)$; q denotes a target, ℓ a real level, and the drift $q = p - \mathcal{E}(p)$ and $\ell = e\,(p - \mathcal{E}(p))$ of (8) do not occur; s_i, c_i, d_i are the scaled values defined below, not those of Section 4.

Let $F(n, \pm)$ be the *fundamental patch* of $\Sigma_n^{\pm}$: the branch $\tau = u \in (0, K)$, $s_2 = +1$. The group G of order eight generated by $(U, V, W) \mapsto (U, V, -W)$ (the branch $\tau = 2K - u$), $(U, -V, W)$ ($\tau = 4K - u$) and $(-U, -V, W)$ ($s_2 = -1$) carries the fundamental patches onto the whole conjugate locus.

Lemma 10.1 (Octants)**.** *On every open sheet $U > 0$. On $F(n, \pm)$, V has the sign $\sigma_n = (-1)^{n+1}$ and W the sign $-\sigma_n$ on $F(n, -)$, $+\sigma_n$ on $F(n, +)$. Hence each fundamental patch lies in one open octant, the $s_2 = -1$ copies lie in $U < 0$ and never meet the $s_2 = +1$ ones, and $F(n, b)$ can meet $g\,F(m, b')$ only for the single $g = (s_V, s_W) \in G$ with*

$$s_V = \sigma_n \sigma_m, \qquad s_W = \epsilon_b \epsilon_{b'} \sigma_n \sigma_m, \qquad \epsilon_- = -1,\ \epsilon_+ = +1.$$

Proof. $U = qD/d > 0$ since $q, D, d > 0$; $V = k f_1 S/d$ with $\operatorname{sgn} f_1 = \sigma_n$ on the cell and $S = \operatorname{sn} u > 0$; $W = -ksC/d$ with $C = \operatorname{cn} u > 0$ and $\operatorname{sgn} \operatorname{sn} p = \sigma_n$ on the left half of the cell, $-\sigma_n$ on the right half. The seams $S = 0$, $C = 0$ have $V = 0$, $W = 0$ and are not in the open patches. □

Proposition 10.2 (Conjugate preimages and the level function)**.** *Let $q = (X, \theta)$, $X \neq 0$, $0 < |\theta| < \pi$, be off the seam of the scalar inverse [11, Proposition 2.1]. Its rotating preimages with $s_2 = +1$ are the points z of its admissible intervals at which the level function*

$$N(z) = \frac{A_q - Q_0}{L(m)}, \qquad L(m) = 4\big(K(m) - E(m)\big), \tag{26}$$

is a nonnegative integer [11]; here $A_q > 0$ is the drift of the target along the axis of its elastica, $L(m)$ the drift per period, and $Q_0 \in (0, L(m))$ the drift of the residual argument $\delta(z) \in (0, 4K)$ (Proposition 10.4 gives Q_0 in closed form). At such a point the source has phase $\psi(z)$, modulus $k(z)$, residual argument $\delta(z)$ and time $t = k\delta + N \cdot 4kK$. Then:

(i) the preimage is a conjugate point if and only if $N'(z) = 0$ [11, Proposition 2.2];

(ii) a conjugate preimage on Σ_n^- has $N = n - 1$ and $\delta > 2K$; one on Σ_n^+ has $N = n$ and $\delta < 2K$;

(iii) q lies on the image of Σ_n^- if and only if N has a critical point at level $n - 1$ with $\delta > 2K$, on the image of Σ_j^+ if and only if N has one at level j with $\delta < 2K$, and on both if and only if both occur.

Proof. (i) is the regularity bridge of the counts paper. (ii) $p = t/(2k) = 2NK + \delta/2$, so $p/(2K) = N + \delta/(4K)$; on the left bracket $p \in ((2n-1)K, 2nK)$, whence $N = n - 1$ and $\delta \in (2K, 4K)$; on the right bracket $p \in (2nK, (2n+1)K)$, whence $N = n$ and $\delta \in (0, 2K)$. (iii) restates (i) and (ii). □

The incidence map. For cells 1–4 the eight fundamental patches were tested against the 31 other patches of these cells (four branches of each bracket, $s_2 = +1$): on 25 slices $k_1 = \text{const}$ of the first patch, up to four nearest sampled approaches within distance 0.35 were refined by least squares on (f_1, f_2, k_2), where f_i is the fractional position in its bracket (with interpolated bracket endpoints), three equations in three unknowns, and accepted at

residual 10^{-10} with interior parameters (`research/e76_incidence_trace.py`). The result (Figure 4): among the 248 ordered pairs the search detected intersections for the following ten unordered pairs

$$F(n,-) \cap g\,F(j,+) \neq \emptyset, \qquad 1 \le n \le j \le 4,$$

each detected from both sides, with g the reflection $W \mapsto -W$ for $n \equiv j$ and $V \mapsto -V$ otherwise, as Lemma 10.1 dictates. No left–left or right–right intersection, pair with $j < n$, or self-intersection was detected. The sampled same-cell traces are consistent with branches from the umbilic toward $k = 1$; this finite search does not establish their global endpoints or exhaust all intersections. At the sampled intersections the left sheet has the larger modulus, and the right-sheet modulus decreases with $j - n$ (from about 0.9 at $j = n$ to about 0.3 at $(1, 4)$). A replay of all 123 saved accepted points recovers target residuals below 6.1×10^{-13}. The finite-grid detector (`research/e76_level_function.py`) finds exactly two critical points, both maxima, at 121 targets; at one further target it finds two maxima and two minima below level zero, and at the remaining target it finds only one maximum. Independent 60-digit differentiation confirms the negative minima and recovers the missed maximum near level 1 at the last target (the detected maximum is near level 2). These tests do not certify completeness of the critical-point search.

Two facts about the level function are theorems. Work in the frame of the heading bisector: let ϕ be the angle of the elastica axis from the bisector, $\omega = 2\phi$, let (A_q, B_q) be the along-axis and transverse coordinates of the target, $R^2 = A_q^2 + B_q^2$, and let ν be the polar angle of the target seen from the axis, $A_q = R\sin(\nu/2)$, $B_q = R\cos(\nu/2)$; along the chart curve of a fixed target ν is the parameter. Write $s_i = k\,\mathrm{sn}$, $c_i = k\,\mathrm{cn}$, $d_i = \mathrm{dn}$ at the initial ($i = 0$) and final ($i = 1$) phase, so that $s_0 = \sin a$, $s_1 = \sin(a - \theta)$ for the axis angle a, and $B_q = c_0 - c_1$.

Lemma 10.3 (The modulus along the chart curve)**.** *Along the chart curve*

$$4m = B_q^2 + 2 - 2\cos\theta\cos\omega + \frac{\sin^2\theta\,\sin^2\omega}{B_q^2},$$

the chart is admissible only if B_q^2 lies strictly between $(1-\cos\theta)(1-\cos\omega)$ and $(1+\cos\theta)(1+\cos\omega)$ (this is $0 < m < 1$; admissibility also requires the sign conditions of [11, Section 2]), and

$$\begin{aligned} B_q\,\partial_\nu m &= A_q\,c_0c_1 + c_0s_1d_1 - c_1s_0d_0, \\ B_q\,\partial_\nu^2 m &= \frac{m(1-m)\,(c_0^3 - c_1^3)}{2\,c_0^2c_1^2} \quad \text{at every critical point of } m. \end{aligned} \tag{27}$$

Hence every critical point of m with $0 < m < 1$ is a strict minimum, and m is unimodal on every admissible interval.

Proof. The transverse coordinate is $B_q = c_0 - c_1$ and both endpoints lie on the circle $s_i^2 + c_i^2 = m$ with $s_0^2 - s_1^2 = \sin\theta\sin\omega$, which gives $c_0 = (B_q - \sin\theta\sin\omega/B_q)/2$ and the closed form; $m = 1$ is $B_q^2 = 4\cos^2(\theta/2)\cos^2\phi$ or $B_q^2 = 4\sin^2(\theta/2)\sin^2\phi$. Along the curve $\partial_\nu A_q = B_q/2$, $\partial_\nu B_q = -A_q/2$, $\partial_\nu s_i = d_i/2$, $\partial_\nu d_i = -s_i/2$, and $\partial_\nu c_0 = (s_1d_1 - s_0d_0 + A_qc_1)/(2B_q)$, $\partial_\nu c_1 = \partial_\nu c_0 + A_q/2$ from $s_0^2 + c_0^2 = s_1^2 + c_1^2$. Differentiating the closed form gives the first identity in (27). Differentiating it once more and substituting the critical value $A_q = s_0d_0/c_0 - s_1d_1/c_1$ gives $2B_q\,\partial_\nu^2 m = c_0c_1(c_0 - c_1) + c_0(1 - 2s_1^2 + s_1^2d_1^2/c_1^2) - c_1(1 - 2s_0^2 + s_0^2d_0^2/c_0^2)$; the identity $(1 - 2s^2)c^2 + s^2d^2 = c^4 + m(1 - m)$ on the circle turns the brackets into $c_i^2 + m(1 - m)/c_i^2$, the cubic terms cancel, and the second identity follows. Its sign is that of B_q, so $\partial_\nu^2 m > 0$ at every critical point with $0 < m < 1$; two strict minima on an open interval would enclose a maximum. □

Proposition 10.4 (Derivative of the level function)**.** *Let $w_i = s_i d_i/c_i$, $A_q^* = w_0 - w_1$, and $\Delta E = E(\varphi_1|m) - E(\varphi_0|m)$ the increment of the incomplete integral of the second kind between the endpoint amplitudes, so that $Q_0 = \delta - \Delta E$. On the chart domain $J = (-\pi/2, \pi/2) \cap (\theta - \pi/2, \theta + \pi/2)$ of the principal axis angle of Lemma 10.5, where $d_0, d_1 > 0$, along the chart curve*

$$L\,\partial_\nu N = -\frac{mB_q}{2c_0c_1} - \frac{\partial_\nu m}{2(1-m)}\Big(A_q^* + \Delta E + 4N\,E(m)\Big). \tag{28}$$

Consequently $\partial_\nu N \neq 0$ at the minimum of m, with the sign of $-B_q c_0 c_1$; the critical points of N are the roots $\Xi = A_q - A_q^*$ of

$$\beta_N \Xi^2 + \big[A_q^* + \Delta E + \beta_N(A_q^* - Q_0)\big]\Xi + \frac{m(1-m)B_q^2}{c_0^2 c_1^2} = 0, \qquad \beta_N = \frac{4E(m)}{L(m)}.$$

Proof. $Q_0 = \mathcal{D}(\varphi_1|m) - \mathcal{D}(\varphi_0|m)$ with $\mathcal{D} = F - E$, $w(\varphi) = \sqrt{1 - m\sin^2\varphi}$, $\partial_\varphi \mathcal{D} = m\sin^2\varphi/w$, $\partial_m \mathcal{D} = [E(\varphi|m) - \sin\varphi\cos\varphi/w]/(2(1-m))$, and the amplitude derivatives $\partial_\nu \varphi_i = (m d_i - s_i \partial_\nu m)/(2m c_i)$. Collecting, the algebraic part of $\partial_\nu A_q - \partial_\nu Q_0$ is $B_q/2 - s_1^2/(2c_1) + s_0^2/(2c_0) = -mB_q/(2c_0c_1)$ and the rest is $\partial_\nu m$ times $-[A_q^* + \Delta E]/(2(1-m))$, using $s_i^3/(c_i d_i)/(2m) + s_i c_i/(2m d_i(1-m)) = w_i/(2(1-m))$; the term $-N\partial_\nu L$ with $\partial_m L = 2E/(1-m)$ completes (28). Multiplying $\partial_\nu N = 0$ by Ξ and eliminating $N = (\Xi + A_q^* - Q_0)/L$ gives the quadratic. The identities are checked numerically in `research/e77_h1_proof_check.py` and `research/e77_Nprime_formula.py`. □

The quadratic makes the critical points of N a double cover of the three-dimensional endpoint data $(m, \varphi_0, \varphi_1)$, so the global structure of the level function can be scanned exhaustively. The scan shows that unimodality at every real level fails near zero heading, and the conjecture is stated in the form the counts need.

A conditional count criterion and its certificate

The critical points of the level function form a double cover of the endpoint data $(m, \varphi_0, \varphi_1)$ (Proposition 10.4), so the sign of N'' at every critical point is a statement about an explicit function of three variables. The proposed interval verification requires a complete replay: the previous run used incorrect incomplete-integral enclosures at negative amplitudes and incorrect Taylor tightening of quadratic roots near coalescence. The repaired primitives pass independent checks, but those checks do not certify the full domain. Write $\mu = (\varphi_0 + \varphi_1)/2$ and $\Delta\varphi = \varphi_1 - \varphi_0$; the statement is invariant under $\mu \mapsto \mu + \pi$ and $\mu \mapsto -\mu$, so $\mu \in [0, \pi/2]$ is a fundamental domain. Let $\mathcal{B}$ be the box

$$\mathcal{B} = \{\, 0.02 \le m \le 0.98,\ 0 \le \mu \le \pi/2,\ 0.02 \le \Delta\varphi \le 2\pi - 0.02 \,\}$$

with five sets removed: the slab $\mathcal{S} = \{|B_q| < 0.02\}$ around the bisector boundary $B_q = 0$ (the seam of the scalar inverse, reached as $\mu \to 0$ and as $\Delta\varphi \to 0, 2\pi$), the two corners $\mathcal{C}_1 = \{\mu < 0.08,\ \Delta\varphi < 0.2\}$ and $\mathcal{C}_2 = \{\mu < 0.08,\ \Delta\varphi > 2\pi - 0.2\}$ near short or full-period residual pieces, and the tube $\mathcal{C}_3 = \{\mu < 0.08,\ |\Delta\varphi - \pi| < 0.1\}$ around the double flip $c_0 = c_1 = 0$, where both roots of the critical quadratic escape to infinity, and the low-modulus set

$$\mathcal{C}_4 = \{0.02 \le m \le 0.06,\ 0.94 \le \mu \le \pi/2,\ 0.41 \le \Delta\varphi \le 0.80\} \cup \{0.78 \le m \le 0.82,\ 1.49 \le \mu \le \pi/2,\ 0.02 \le \Delta\varphi$$

the union of the 32 grid cells (a wedge of 31 cells and one further cell) in which the budget passes of the interval engine leave boxes open. On the wedge the small critical root is an

$O(m)$ difference of $O(1)$ quantities and the interval enclosures widen faster than the margin of (29) shrinks; a 50-digit pointwise survey finds $mN''(\nu_*) \to -c(\mu, \Delta\varphi)$ with $c \geq 0.0013$ on the wedge, so the first alternative holds there numerically with a margin growing like $1/m$, but this is not certified and $\mathcal{C}_4$ is declared as an exclusion. By *the family* we mean below the grid of $100 \times 120 \times 240$ endpoint data $(m, \mu, \Delta\varphi)$ of `research/e78_statement_L.py`, which samples the ambient endpoint domain, including the removed sets, corners and tube; on this finite grid the tube carries no interior minimum, its critical points having $R \geq 0.4$ and $N \geq 0.07$.

Certificate. The computation establishes, at every interior critical point ν_* with $A_q > 0$ and endpoint data in $\mathcal{B} \setminus (\mathcal{S} \cup \mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{C}_3 \cup \mathcal{C}_4)$, the predicate

$$N''(\nu_*) < 0 \qquad \text{or} \qquad N(\nu_*) < \tfrac{1}{2} \ \text{ and } \ R < L\big(m(\nu_*)\big), \tag{29}$$

where $R = \sqrt{A_q^2 + B_q^2}$. The run of 21 September 2026 described in the numerical paragraph below, with the repaired primitives, closes every one of the 38,400 cells outside the five sets: the 304 boxes left open by its budget passes (total volume 0.049% of $\mathcal{B}$) all lie in $\mathcal{C}_4$, 285 in the wedge and 19 in the further cell, and are listed in `data/e80_full_grid_open_cells.json`. The excluded sets occupy 2.56% of $\mathcal{B}$ (slab 2.06%, corners 0.25%, tube 0.14%, $\mathcal{C}_4$ 0.11%). The corollary below is an analytic implication of the predicate; off $\mathcal{S} \cup \mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{C}_3 \cup \mathcal{C}_4$ its hypothesis is the certified statement.

Lemma 10.5 (Compactness of positive superlevels; [11, Proposition 2.4])**.** *Fix $q = (X, \theta)$ with $X = (x, y) \neq 0$, $0 < |\theta| < \pi$ and $x\cos(\theta/2) + y\sin(\theta/2) \neq 0$. Let I be a maximal admissible component of its scalar inverse. For every $\ell > 0$, $\{a \in I : N(a) \geq \ell\}$ is a compact subset of I, possibly empty.*

Proof. Use the principal axis angle a, with $b = a - \theta$, on $J = (-\pi/2, \pi/2) \cap (\theta - \pi/2, \theta + \pi/2)$, and write $A_q = x\sin a - y\cos a$, $B_q = x\cos a + y\sin a$. The circle equations give

$$m = \sin^2 a + c_0^2 = \sin^2 b + c_1^2, \qquad \sin\theta\sin(2a - \theta) = -B_q(c_0 + c_1).$$

At least one of $|a|, |b|$ is at least $|\theta|/2$, so $m \geq m_h := \sin^2(\theta/2) > 0$. The defining integral makes L strictly increasing, with limits 0 and $+\infty$ at 0 and 1 [13, Sections 19.2, 19.12]. On a nonempty positive superlevel,

$$\ell L(m) \leq A_q - Q_0 < A_q \leq R, \qquad m_h \leq m < M_\ell := L^{-1}(R/\ell) < 1, \qquad A_q > \ell L(m_h) > 0.$$

Take a sequence in this superlevel converging in $\overline{J}$. A limit at an end of J would force $m \geq 1$, since a or b would equal $\pm\pi/2$. At an interior limit, A_q stays positive. If $B_q \to 0$, boundedness of c_i and the second circle identity would give $\sin(2a - \theta) = 0$. On $\overline{J}$ one has $-\pi < 2a - \theta < \pi$, hence $a = \theta/2$, contradicting the target's off-seam hypothesis. Thus $B_q \neq 0$ at the limit, and the algebraic inverse extends there with $0 < m < 1$. Local phase lifts make Q_0 continuous: the two endpoint phases cannot coincide modulo $4K$, since their principal angles differ by $\theta \neq 0$. Hence the limit is admissible with $N \geq \ell$. A component is closed relative to the admissible domain, so the limit belongs to I. This proves compactness; the parameters $z = \tan a$ and ν give the same conclusion. □

Corollary 10.6 (Counts at positive levels)**.** *Let q and its maximal admissible component I satisfy Lemma 10.5. Assume that every critical point of N with $N \geq \frac{1}{2}$ and every interior local minimum of N satisfies* (29)*. Off $\mathcal{S} \cup \mathcal{C}_1 \cup \mathcal{C}_2 \cup \mathcal{C}_3 \cup \mathcal{C}_4$ this is the certified statement; on those sets it is an assumption. Then every critical point of N on I at a level $\geq \frac{1}{2}$ is*

a strict maximum, at most one critical point of N on I lies at a level ≥ 1, and for every real $\ell \geq 1$ the superlevel set $\{\nu \in I : N(\nu) \geq \ell\}$ is an interval (possibly empty). Writing $M = \sup_I N$, for every real $\ell \geq 1$ the number of distinct solutions of $N = \ell$ is 2, 1 or 0 according as $\ell < M$, $\ell = M$ or $\ell > M$. In particular the count for integer $n \geq 1$ is at most two and non-increasing in n.

Proof. By (29) a critical point with $N \geq \frac{1}{2}$ has $N'' < 0$ and is a strict maximum, and an interior local minimum has $N < \frac{1}{2}$ and $R < L(m)$. Let $a < b$ lie in I with $N(a), N(b) \geq \ell \geq 1$. If N dips below ℓ between them, its minimum on $[a, b]$ occurs at an interior point d. By Lemma 10.3, $m(d) \leq \max(m(a), m(b)) =: m_{\max}$. At either endpoint $j \in \{a, b\}$, $L(m(j)) \leq N(j)L(m(j)) = A_q(j) - Q_0(j) \leq R$. Consequently $R \geq L(m_{\max}) \geq L(m(d))$, contradicting the predicate at d. Every superlevel set at a level $\ell \geq 1$ is therefore an interval, possibly empty. Two strict maxima with values at least 1 would enclose an interior local minimum; the same inequality, using those maxima as a, b, excludes it. Hence there is at most one such critical point. If $N = \ell \geq 1$ had three distinct solutions $a < b < c$, superlevel connectedness would give $N \geq \ell$ on $[a, c]$, making b a local minimum, a contradiction. This proves the two-solution bound without assuming the existence of a maximum. Now $M \leq R/L(\sin^2(\theta/2))$ is finite. If $M > 0$, a maximizing sequence eventually lies in the compact superlevel at $M/2$, so M is attained. For $1 \leq \ell < M$, compactness of the superlevel and continuity give a crossing on each side of the maximum; the two-solution bound excludes any more. At $\ell = M \geq 1$ there is exactly one maximizer, by the critical-point conclusion, and above M there are no roots. If $M < 1$ there are no positive integer roots. This proves the stated count law. The inequality at a, b needs $N \geq 1$; it does not exclude two maxima whose values lie in $[\frac{1}{2}, 1)$. □

Remark 10.7. The maximal-component and target off-seam hypotheses are essential: a truncated interval or a chart component of a seam target can have a positive level at an excluded end. The monotonicity conclusion starts at $n = 1$. It does not extend from $n = 0$ to $n = 1$: in the far-field regime of [11, Theorem 3.3], one component has one level-zero root and two roots at level one once its maximum exceeds one. The level-zero clauses therefore remain separate. The corollary requires the endpoint-data hypothesis at all of its specified critical points; membership of a target in a spatial region alone does not establish that hypothesis. Some necessary constraints are explicit: along a component $m \geq \sin^2(\theta/2)$, so $m \geq 0.02$ when $|\theta| \geq 0.284$; and a critical point with $m > 0.98$ and $N \geq \frac{1}{2}$ needs $R > L(0.98)/2$, since $NL = A_q - Q_0 \leq R$. On $m \leq 0.98$, an amplitude increment of size ϵ gives an elliptic phase increment between ϵ and $\epsilon/\sqrt{1 - 0.98}$, by the integral defining F. Thus $\Delta\varphi$ within 0.02 of 0 or 2π puts the phase increment within $0.02/\sqrt{0.02} < 0.142$ of zero or one full period, respectively. The slab $\mathcal{S}$ is a neighbourhood of the seam $B_q = 0$, off which Proposition 10.2 is stated. The corner and tube samples do not prove exclusions on those continuous sets. On the finite family the reported largest values at an interior minimum are approximately 0.45 for $R/L(m)$ and 0.39 for N on the finest grid of `research/e78_statement_L.py`, and up to 0.43 for both on a coarser grid, so the margin to the threshold $\frac{1}{2}$ of (29) is about 0.05; these are numerical observations, not uniform bounds. Proposition 10.10 separately covers fixed far-field rays. Level 0 is outside the positive-level corollary.

Method. Every quantity of Proposition 10.4 is evaluated in ball arithmetic [9, 17] (`arb`, through `python-flint`) as a first-order Taylor model over a box: centre value, box enclosure, and box enclosure of the gradient with respect to $(m, \mu, \Delta\varphi)$, the enclosure being tightened by the mean-value form after every operation. The incomplete integrals are enclosed by monotonicity: F and E increase in φ, and in m the derivatives $\partial_m F = \frac{1}{2}\int_0^\varphi \sin^2\theta\,(1 -$

$m\sin^2\theta)^{-3/2}d\theta$ and $\partial_m E = -\frac{1}{2}\int_0^{\varphi}\sin^2\theta\,(1-m\sin^2\theta)^{-1/2}d\theta$ have the signs of φ and of $-\varphi$, so the extremes of each over a box are corner values, and the union of the four corner evaluations is the enclosure for either sign of φ (the lower endpoint phase $\mu - \Delta\varphi/2$ is negative on most of $\mathcal{B}$); the derivatives are in closed form. Two reformulations make the evaluation regular: with $\Gamma = c_1 s_0 d_0 - c_0 s_1 d_1$, the polynomial $\Lambda = (\Gamma^2 - m(1-m)B_q^2)/(c_0 c_1)$ and $\mathcal{W} = \Delta E + \beta_N(A_q - Q_0)$, the critical condition is the quadratic

$$\Psi(A_q) = -c_0 c_1 \beta_N A_q^2 + bA_q + c = 0, \qquad b = -\Gamma - c_0 c_1(\Delta E - \beta_N Q_0) + \beta_N \Gamma,$$

$$c = \Lambda + \Gamma(\Delta E - \beta_N Q_0),$$

in A_q, with $\Psi = 2B_q(1-m)L(m)\,\partial_\nu N = \Lambda - A_q\Gamma - (c_0 c_1 A_q - \Gamma)\mathcal{W}$, so that at a root $\operatorname{sgn} N'' = \operatorname{sgn}(B_q\,\partial_\nu\Psi)$. For the root test a polynomial form avoids division at an endpoint flip. Put $\mathcal{F} = c_0 c_1 A_q - \Gamma = B_q\,\partial_\nu m$ and $\mathcal{W} = \Delta E + 4E(m)N$. At $\Psi = 0$, hence $\partial_\nu N = 0$,

$$B_q\,\partial_\nu \mathcal{W} = \frac{d_1 R_1 - d_0 R_0 - \mathcal{F} A_q}{2m}, \qquad \partial_\nu \varphi_i = \frac{R_i}{2mB_q},$$

where the R_i are the polynomial amplitude numerators of `research/e78_regular_prep.py`. Substituting this expression when differentiating Ψ gives a cubic $P_{\text{crit}}(A_q) = p_3 A_q^3 + p_2 A_q^2 + p_1 A_q + p_0$ with polynomial coefficients in $(s_i, c_i, d_i, \Delta E, \beta_N, Q_0)$ and

$$2mB_q\,\partial_\nu\Psi = P_{\text{crit}}(A_q) - \frac{m\beta_N\mathcal{F}}{1-m}\Psi(A_q).$$

This identity follows also from the omitted term $4E(m)\partial_\nu N$ in $\partial_\nu\mathcal{W}$ and is checked exactly modulo the endpoint circle identities. Thus P_{crit} equals $2mB_q\,\partial_\nu\Psi$ at every critical root, including flips by continuity, but is not the unrestricted derivative away from those roots. Its leading coefficient is

$$p_3 = c_0^2 c_1^2 - m\beta_N(c_0^2 + c_1^2).$$

The unrestricted cubic in `research/e78_cubic_prep.py` includes $2m\beta_N'(m) = -1 - m\beta_N^2/(1-m)$ and has the additional leading term $m\beta_N^2 c_0^2 c_1^2/(1-m)$. The earlier polynomial is valid for the critical-root test; interpreting it as the unrestricted derivative was incorrect. The verifier uses the polynomial coefficients generated by `research/e80_critical_cubic_prep.py`. When $p_3 < 0$ on a box, the *cubic threshold*

$$A_{\text{th}} = 1 + \frac{|p_2| + |p_1| + |p_0|}{-p_3},$$

formed from the bounds of the enclosures, gives $B_q\,\partial_\nu\Psi < 0$, hence $N'' < 0$, at every root $A_q \geq A_{\text{th}}$: for $A_q \geq 1$, the sum of the lower-order terms is at most $(|p_2| + |p_1| + |p_0|)A_q^2$, while $(-p_3)A_q > |p_2| + |p_1| + |p_0|$. When the enclosure and threshold comparisons succeed, this settles the root that escapes to infinity at a flip; otherwise the box remains undecided. Let $\mathcal{E} = \{N < \frac{1}{2},\ R < L(m)\}$ be the *escape set*, the second alternative of (29). A box passes when its enclosures establish that Ψ has no real root, or that every root A_q of Ψ on the box satisfies

$$\begin{aligned} A_q \leq 0, \qquad &\text{or} \qquad B_q\,\partial_\nu\Psi < 0, \qquad \text{or} \qquad A_q \geq A_{\text{th}}, \\ &\text{or} \qquad N < \tfrac{1}{2} \ \text{ and } \ R < L(m) \ \text{ at the root} \end{aligned} \tag{30}$$

(a root that escapes to infinity because the leading coefficient of Ψ straddles zero passes when its modulus is at least A_{th}; when the enclosure of the discriminant of Ψ straddles zero, its square root has no first-order model and is taken as the plain interval $[0, \sqrt{\max \text{disc}}]$, the

roots and every quantity computed from them then being carried as plain intervals without mean-value tightening); the first alternative is an inadmissible root, the second and third give $N'' < 0$, the fourth is the escape set, so a passed box satisfies (29). Otherwise the box is bisected along its widest side, down to width $\min(2 \cdot 10^{-3}, m/25)$ in the first pass and $2.5 \cdot 10^{-4}$ in the refinement. The removal of the slab $\mathcal{S}$ and of $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3$ is implemented by passing without evaluation every box that lies inside one of them (the verifier records them as deferred); these sets are outside the scope of the proposed certificate. The fifth set $\mathcal{C}_4$ is declared after the run as the union of the cells whose budget passes left boxes open; its cells were evaluated like all others, their accepted leaves are kept, and no box outside $\mathcal{C}_4$ remains open. The certificate run covers the initial grid of $24 \times 20 \times 80$ cells, one checkpoint per cell, with four budget passes of 256, 4,096, 16,384 and 65,536 evaluations; a box is accepted by the first of four reviewed predicate methods (the scaled critical-root test above and three auxiliary root-branch tests; `research/e80_combined_predicate.py`), bisected along its widest side otherwise, and left open when the budget of its cell is exhausted. Boxes inside the four deferred sets are recorded as deferred, not counted as verified, and are outside the scope of the certificate; no box reached the minimum width. The 38,400 checkpoints carry the source hashes and the canonical subdivision trace of their cell; an independent semantic replay re-proves every accepted leaf, re-walks every partition and checks each resumed pass against its parent, with no failure: 1,072,966 accepted leaves, 15,777 deferred leaves, 304 open leaves (all in $\mathcal{C}_4$) (`data/e80_full_grid_pass1.json`, `data/e80_full_grid_audit.json`, `data/e80_full_grid_open_cells.json`). The old run statistics and residual clusters of the defective verifier are withdrawn as certificate evidence and retained only in the research notebook; volume fractions use the ambient rectangle before exclusions as denominator. Scripts: `research/e78_verify5.py`, `research/e78_tm.py`, `research/e78_regular_prep.py`, `research/e78_cubic_prep.py`, `research/e80_combined_predicate.py`, `research/e80_combined_checkpoint.py`, `research/e80_full_grid_local.py`, `research/e80_full_grid_audit.py`.

Conjecture 10.8 (Two maxima)**.** *For every target off the seams with two admissible intervals and every integer $n \geq 0$, the equation $N = n$ has at most two solutions on each interval; for $n \geq 1$ their number is non-increasing in n, and at levels ≥ 1 the level function has on each interval at most one critical point, a maximum; at level* 0 *no critical point other than the maximum of the interval lies on the level, so that the conjugate preimages of a target are its interval maxima at integer levels; the two maxima have different residual types, $\delta > 2K$ at one and $\delta < 2K$ at the other; and the maximum of right type exceeds the maximum of left type, $M_R > M_L$. Corollary 10.6 proves the bound of two solutions, uniqueness of a critical point, and count monotonicity for $n \geq 1$ on maximal off-seam components under its explicit sign-predicate assumption. The repaired verifier has not yet established that assumption on a certified domain. The unconditional positive-level claims, the level* 0 *(its count and critical-point clause), the removed sets and margins of $\mathcal{B}$, and the two clauses on the types remain conjectural.*

Evidence. A replay with 1,000 accepted random targets in $0.5 < |X| < 14$, $0.05 < |\theta| < \pi - 0.05$ and at least one finite admissible interval gives 1,798 finite intervals: the sampled detector reports one nonnegative-level maximum on 1,667 and none on 131. A separate sample of 600 targets with two finite intervals gives 594 with both interior maxima detected; their types differ, and $M_R - M_L$ ranges from 0.1396 to 274.913, below 1 on 94 targets (`research/e76_unimodality_scan.py`, seed 20260907, argument 1000; `research/e76_two_maxima.py`, seed 7, argument 600). These finite-grid results do not exclude missed critical points.

A historical $60 \times 120 \times 120$ endpoint-data scan reported 405,558 quadratic roots with $A_q > 0$, $N \geq 0$: 402,796 maxima and 2,762 minima. A portable replay using a separately derived critical-curvature formula reproduces these finite-grid counts with both the original and a cancellation-resistant quadratic formula (`research/e80_e77_replay.py`); all 2,762 candidate targets match the saved survey in order, with spatial-radius error below 1.9×10^{-12}. Selected targets also pass independent 70-digit differentiation checks. The original scratch polynomial remains unavailable; this replacement does not give an interval certificate or certify every extremum detected by the subsequent chart scans. The saved survey contains 2,762 records marked as minima by those scans. Their $|\theta|$ is at most 0.1597, $|X|$ at most 1.482, and minimum level at most 0.2743; the smaller recorded neighboring maximum is at most 0.2744. No record flags four crossings of an integer level. These saved chart observations do not certify every detected stationary point or exclude additional unsampled extrema. The finite-difference detector is unreliable for the shallowest dips: on six reconstructed targets with 5–11 historically reported critical points, 30 of 81 examined sampled derivative signs are incorrect in binary64, while 70-digit recomputation of the same stencils removes those mismatches. Local interval checks isolate two maxima and one minimum for each of these six targets, all at levels strictly between 0 and 1; they do not establish completeness outside the 18 isolating boxes (`references/review_2026-09-21_neighbors/`). The finer grid discussed after Corollary 10.6 reports levels up to 0.39. A separate saved small-heading survey records 86 dips, none flagged as crossing an integer level four times (`research/e77_critical_quadratic.py`, `research/e77_dips_survey.py`, `research/e77_dips_neighbors.py`, `data/e77_dips_survey.json`). The locally certified positive-level minima rule out all-real-level unimodality; the integer statement remains conjectural.

Corollary 10.9 (Conditional on Conjecture 10.8)**.** *Let $q = (X, \theta)$ satisfy $X \neq 0$, $0 < |\theta| < \pi$, be off the scalar-inverse seams, and have two admissible intervals as in the conjecture. Such a target has at most two rotating conjugate preimages. Within this target domain the left-sheet images are pairwise disjoint, and so are the right-sheet images. For the octant-matching g, with the intersections restricted to this domain,*

$$F(n,-) \cap g\,F(j,+) = \{q :\ M_L(q) = n-1,\ M_R(q) = j\},$$

which is empty unless $j \geq n$. The number of rotating geodesics through a target is determined by $\lfloor M_L \rfloor$, $\lfloor M_R \rfloor$ and the end values of N on the two intervals; it is constant on the chambers cut by the sheets and changes by two across a sheet.

Proof. By Proposition 10.2 conjugate preimages are critical points at integer levels; with one maximum per interval at levels ≥ 1 and no critical point at level 0 other than the interval maxima, there are at most two, of different types, and a target on $\Sigma_n^- \cap \Sigma_{n'}^-$ would need $M_L = n-1 = n'-1$. The incidence set is the double integer condition, and $M_R > M_L$ forces $j > n-1$. With at most two solutions per integer level, non-increasing at positive levels, the number of integer crossings is read off from the maximum and the end values, and crossing a sheet changes the count by two (a maximum passing an integer level creates or destroys a pair). □

In the far field the unimodality part is a theorem at every real level.

Proposition 10.10 (Far-field incidences)**.** *Fix a spatial direction a_0 and a heading $\theta \neq 0$ with $b_0 = a_0 - \theta$, both in $(-\pi/2, \pi/2)$ and with $\sin^2 b_0 \neq \sin^2 a_0$, and let $q_R = (R\,e(a_0), \theta)$ with $e(a_0) = (\sin a_0, -\cos a_0)$ and $L_* = L(\max(\sin^2 a_0, \sin^2 b_0))$ as in [11, Section 3]. There is R_0 such that for $R \geq R_0$: the level function of q_R has exactly one critical point on each of its*

two admissible components, a nondegenerate maximum, of heights $M_\pm(R) = R/L_* + O(1)$; hence q_R has at most two conjugate preimages, it lies on the conjugate locus exactly when $M_-(R)$ or $M_+(R)$ is an integer, on two sheets exactly when both are, and the two sheets through such a point belong to cells whose indices differ by $O(1)$ uniformly in R.

Proof. The first assertion is [11, Theorem 3.3], and the exclusion of the sign $s_2 = -1$ at large R is [11, Section 3]; the rest is Proposition 10.2, the level of a conjugate preimage being $n-1$ or n for a sheet of cell n. □

What is proved here is therefore the octant restriction, the level-function characterization, the conditional count implication at levels ≥ 1 (Corollary 10.6), and the far-field structure along fixed rays; the map for the first four cells is computed numerically only; the global structure at all radii, and the inequality $M_R > M_L$ behind the rule $j \geq n$, rest on Conjecture 10.8.

11 Numerical verification

The accompanying numerical scripts are in the project repository, folder `research/`, prefixes `e74`–`e78`. The historical summaries below describe finite-sample diagnostics; the complete collection of these runs has not been independently replayed. At 84 points of the open sheet, six moduli including 0.9089 (near, but not equal to, k_0) and 0.99, both branches, the independently replayed finite-difference differential of (3) at 40 digits has singular values $\sigma_2/\sigma_1 \geq 4.9 \times 10^{-3}$, $\sigma_3/\sigma_1 \leq 6 \times 10^{-28}$, kernel time component $\leq 7 \times 10^{-25}$, and the kernel derivative of $\det d\,\mathrm{Exp}$ has the sign predicted by (14) at every point. The replayed minimum of σ_2/σ_1 is $0.004990276288\ldots$, correcting the previously rounded lower bound. Independent 60-digit differentiation at eight selected points, including both phases near this minimum, agrees with separately integrated Hamiltonian and variational equations: the largest endpoint error is below 9×10^{-14}, and the largest differential error, normalized by $1 + \max |D\,\mathrm{Exp}|$, is below 4×10^{-13} (`research/e80_geometry_audit.py`, `references/review_2026-09-21_geometry_diagnostics/`). These are finite-sample diagnostics, not interval enclosures. The remaining historical summaries have not all been independently replayed. The symbolic θ_k, θ_ψ, $v(J_2)$ agree with finite differences to 3×10^{-20}. At the Maxwell edge and at $(2K, K)$, for four and five moduli, $|v(J_2)| \leq 3 \times 10^{-28}$, away from k_0 the second kernel derivatives agree with Theorem 6.1 to ten and twelve digits and the rank condition holds; at k_0 the singular values are 1, 8×10^{-22}, 0. On the α_1 edge $Q \in [9 \times 10^{-6}, 8 \times 10^{-3}]$ and $|\mathsf{Z}| \leq 3 \times 10^{-42}$ at four moduli. The atlas chart agrees with the raw map to 2×10^{-28} at 250 points; $H_p \geq 0.0239$, $a' \geq 0.0465$, $\zeta' \geq 0.003$ there; H_1 increases on 92 moduli from 0.02 to 19.6; U increases along the slope levels $H = 0.05, 0.3, 1, 3, 10$; the lower seam is ordered; and the escape bound holds on the strips $k = 0.9, 0.99, 0.999, 0.999999$, where the minimal spatial radius rises from 3.26 to 14.18. Independent replay covers the edge and chart samples. A separate 50-digit implementation checks the upper slopes, lower seams, five nonempty slope levels and four escape strips, with selected 60-digit raw-map derivative checks (`references/review_2026-09-21_edge_embedding/`). All bounds remain finite-sample diagnostics. On the later sheets, at 180 points of cells 1–5, both brackets and both branches, three moduli including 0.9089, the identity (14) holds to 10^{-8} and the corank is one; nQ divided by the leading term of (25) tends to one with an $O(1/n)$ deviation up to $n = 80$; the seam script tests 30 pairs (k, n) and prints ten of them, reporting agreement with (24) to 10^{-18} and $d(kb_n)/dk > 0$. The derivative $\partial_k^{\mathrm{src}} \beta_1$ instead has sign $(-1)^n$, by (24); independent 60-digit checks at $k = 0.6, 0.95$ and $n = 1, 2, 3$ confirm this sign and the formula. The intrinsic umbilic test gives D_4^+ at the central points of

cells 1–3 (`research/e75_all_sheets.py`, `research/e75_step2_seams_all_n.py`). These later-sheet scripts and the 48 leading-term samples were replayed; a separate 60-digit implementation checks all 180 open-sheet points, critical Hessians at cells $1, 2, 3, 4, 6$ and six gap examples (`references/review_2026-09-21_later_sheets/`).

The independent September 2026 review checked the exact fold certificates, atlas identities and umbilic jets, replayed the saved incidence targets and the two random-target summaries above, and checked selected raw endpoints and the scalar-inverse dependency at 60 digits. Its logs, source hashes and limitations are retained in `references/review_2026-09-20_caustic/` and `references/review_2026-09-21_caustic_provenance/`. The interval sign predicate still requires a complete corrected replay and an independent coverage and residual audit.

12 Relation to prior work

Sachkov [14] proves that the first conjugate time lies in $[2kp_0, 4kK]$, states the bracket $[2kp_0, 2kp_1]$ and gives the Jacobian (4); Moiseev–Sachkov [12] give the exponential map (3) and the half-angle chart; [10] makes the first conjugate time exact and supplies the Riccati identities (9). The Lagrangian classification, with the determinacy and versality of the generating families, is Arnold's [3]; the fold and cusp criteria are Morin's [6, Chapter VII]; Itoh–Kiyohara [8] describe the conjugate locus of an ellipsoid with the same three local types. The original derivations of the atlas, the univalence argument and the closed-incidence argument were made in the half-angle chart; the present paper re-derives the corank, kernel, fold identity and edge classification in raw coordinates by the first variation, which shortens them and makes the fold factor explicit. The local structure of every later sheet is Section 9, whose certificates and seam inequalities are reproduced in Appendix A, and the incidences between sheets are Section 10; the multiplicity of every point follows from Conjecture 10.8 and is the subject of [11].

Code and data availability

All scripts, the certificate tables and the figure code are in the repository cited above: `research/e74_p8_kernel_rederivation.py`, `research/e74_step2_fold_identity.py`, `research/e74_ring.py`, `research/e74_step3_edges.py`, `research/e74_step4_d4.py`, `research/e74_step5_embedding_audit.py`, `scripts/caustic_figures.py`, `scripts/review/verify_q_certificate_fractions.py`, `scripts/first_caustic_projection_reduction.py`, `scripts/universal_right_fold_proof.py` (run with `PYTHONPATH=.:src`, as it imports the `scripts` package), `research/e78_verify5.py`, `research/e78_tm.py`, `research/e78_regular_prep.py`, `research/e78_cubic_prep.py`, `research/e78_statement_L.py`, `research/e75_leading_term.py`, `research/e75_step3_certificate_domains.py`, `scripts/first_caustic_global_univalence_proof.py`, `research/e79_atlas_identity.py`. The certificate table `data/open_sheet_q_certificate.json` records the SHA-256 digest `7e9fabf7…950c` of its canonical coefficient table in its `sha256` field, and `data/universal_right_sv_certificates.json` likewise `3643ff00…13ff07` (digests of the tables, not of the files). The prior interval summaries `data/e78_verify_L.json` and `data/e78_refine_L.json` are historical outputs of the defective verifier, not proof evidence. Independent primitive regression checks are recorded in `data/e80_primitives_check.json`; the regenerated certificate and its replay are `data/e80_full_grid_pass1.json`, `data/e80_full_grid_audit.json` and `data/e80_full_grid_open_cells.json`. The long working manuscript `manuscripts/`

`first_conjugate_time_c2.tex` of the repository (the research compendium) contains the original derivations in the half-angle chart; the three proofs of Appendix A are transplanted from it.

AI assistance

This paper was prepared with the assistance of Claude (Anthropic), a large language model used through the Claude Code agent in September 2026. The model re-derived the kernel, the fold identity and the edge classification in raw coordinates from the author's working manuscript, found the factor in (14) numerically before proving it, wrote the verification and figure code, and drafted the text. Every statement was checked by the author, who takes full responsibility for the content; no claim rests on the model's output alone. OpenAI Codex assisted with the September 2026 adversarial review, independent certificate and endpoint checks, and the critical-root polynomial reformulation.

A Three proofs from the working manuscript

The three results below are used in Proposition 7.1, Lemma 8.2 and Theorem 9.1. They are transplanted, in the notation of this paper, from the working manuscript named in the code and data section.

Proposition A.1 (Reduction of the projection determinant)**.** *In the amplitude chart* $(A, B, k) = (\operatorname{am} p, \operatorname{am} \tau, k)$ *of the source put*

$$\Delta = \det \frac{\partial(U, V, J_2)}{\partial(A, B, k)}.$$

On the fundamental patch $0 < B < \pi/2$, $(J_2)_B = 2 \sin B \cos B \, (\alpha - \beta) > 0$, *the restriction of* (U, V) *to the surface* $J_2 = 0$ *has*

$$\det \frac{\partial(U, V)}{\partial(A, k)}\Big|_{J_2=0} = -\frac{\Delta}{(J_2)_B}, \tag{31}$$

and on $J_2 = 0$, *in the variables* $g = -c$, $\mathsf{x} = \mathsf{u}/g$, $\mathsf{t} = L_0\mathsf{v}/\mathsf{u}$, $\mathsf{y} = g^2$ *of Theorem 5.1,*

$$\Delta = \Pi \cdot g^3 \mathsf{x}^2 \left[L_0^2 + L_0 \mathsf{y}(1 - \mathsf{t} + \mathsf{tx}) + L_0 \mathsf{t} + \mathsf{ytx} \right] P(L_0, \mathsf{x}, \mathsf{t}, \mathsf{y}), \tag{32}$$

with P *the certificate polynomial of* (17) *and* Π *a product of factors of fixed sign on the patch. Consequently* $\partial(U, V)/\partial(z, k) \neq 0$ *on every open patch, and the sign in* (21) *is constant on the fundamental patch.*

Proof. Expanding Δ along its third row after the column operations that kill $(J_2)_A$ and $(J_2)_k$ gives (31), because on $J_2 = 0$ one has $B_A = -(J_2)_A/(J_2)_B$ and $B_k = -(J_2)_k/(J_2)_B$. The reduction (32) is an exact computation in the ring of Section 4: Δ is formed from (20), (4) and the modulus derivatives of Theorem 4.1; the factor $C = \cos B > 0$ is divided out; $J_2 = 0$ is imposed through $S^2 = r$; the Riccati variables enter by $q = -ds/(a + c)$, $e = ds/(b + c)$ (that is, (11) and its analogue for b); the result is reduced modulo $c^2 + s^2 = 1$, $d^2 + ms^2 = 1$; the fixed factor $a\,d\,k^2 s^7 (k^2 - 1)(ks - 1)^4 (ks + 1)^4$ is removed; and the substitution $c = -g$, $a = g(1 - \mathsf{x})$, $b = g(1 + \mathsf{xt}/L_0)$, $m = 1/(1 + L_0)$, $s^2 = 1 - g^2$ leaves $g^3\mathsf{x}^2$ times the bracket of (32) times a polynomial in $(L_0, \mathsf{x}, \mathsf{t}, g^2)$, which is verified to coincide with P term by term. The script `scripts/first_caustic_projection_reduction.py` carries out this chain in exact arithmetic and raises an error if the final polynomial differs

from P. The bracket is positive because $0 < \mathsf{t} < 1$, and $P > 0$ by Theorem 5.1; so $\Delta \neq 0$, hence $\partial(U,V)/\partial(A,k) \neq 0$ on $J_2 = 0$ by (31), and $\partial(U,V)/\partial(z,k)$ differs from it by the factor $\partial A/\partial z = \partial_z \operatorname{am}(\mathcal{P}(z,k),k) > 0$, since $\mathcal{P}$ increases with z by the first-time theorem. The other patches follow by the sign actions of Proposition 7.1. □

Lemma A.2 (Three bounds at the supercritical endpoint)**.** *Let $k_0 < k < 1$, $m = k^2$, $w(\theta) = \sqrt{1 - m\sin^2\theta}$, and let $\varpi = 2K - p_\alpha \in (0,K)$, $\tilde e = 2E - \mathcal{E}(\varpi)$ be as in the proof of Lemma 8.2. Then*

$$\textit{(i)}\ k_0^2 > \tfrac34, \qquad \textit{(ii)}\ \tilde e > \tfrac32, \qquad \textit{(iii)}\ 2(1-m)(K-E) < \tfrac34.$$

Proof. (i) Pair θ with $\pi/2 - \theta$ in the complete integrals. From $w(\theta)^2 w(\pi/2-\theta)^2 = (1 - m\sin^2\theta)(1 - m\cos^2\theta) = 1 - m + m^2\sin^2\theta\cos^2\theta \geq 1 - m$,

$$K = \int_0^{\pi/4} \frac{w(\theta) + w(\pi/2-\theta)}{w(\theta)\,w(\pi/2-\theta)}\,d\theta < \frac{1}{\sqrt{1-m}} \int_0^{\pi/4} \big[w(\theta) + w(\pi/2-\theta)\big]d\theta = \frac{E}{\sqrt{1-m}}$$

for every $0 < m < 1$. At $m = 3/4$ this is $2E > K$. Since $2E - K$ decreases strictly in m (E decreases and K increases) and vanishes exactly at $m = k_0^2$, we get $k_0^2 > 3/4$.

(ii) With $\tilde s = \operatorname{sn}\varpi$, $\tilde g = \operatorname{cn}\varpi$, $\tilde d = \operatorname{dn}\varpi$, the endpoint equation $\alpha_1(p_\alpha) = 0$ reads, as in the proof of Lemma 8.2, $q(\tilde e\tilde s - \tilde g\tilde d) + \tilde d(\tilde g\tilde e + \tilde s\tilde d) = 0$ with $q, \tilde d, \tilde g, \tilde e, \tilde s > 0$, so $D_0(\varpi) < 0$ for $D_0(x) = e(x)\operatorname{sn}x - \operatorname{cn}x\operatorname{dn}x$, $e(x) = 2E - \mathcal{E}(x)$. Differentiating with $e' = -\operatorname{dn}^2$ gives $D_0'(x) = \operatorname{cn}x\,\big[e(x)\operatorname{dn}x + m\operatorname{sn}x\operatorname{cn}x\big] > 0$ on $(0,K)$. Let $\phi_0 = \pi/6$ and $x_0 = F(\phi_0,k)$, so that $D_0(x_0) = \frac12\big[2E - E(\phi_0\,|\,m) - \sqrt3\,w(\phi_0)\big]$. Put $h(\theta) = w(\theta) - \cos\theta$; then $h(0) = 0$ and $h'(\theta) = \sin\theta\,(w - m\cos\theta)/w > 0$ because $w^2 - m^2\cos^2\theta = (1-m)(1+m\cos^2\theta) > 0$, so h increases and is positive on $(0,\pi/2]$. Subtracting the same expression at $m = 1$, where it equals $2 - \sin\phi_0 - \sqrt3\cos\phi_0 = 0$,

$$2E - E(\phi_0\,|\,m) - \sqrt3\,w(\phi_0) = \int_0^{\phi_0} h + 2\int_{\phi_0}^{\pi/2} h - \sqrt3\,h(\phi_0) > \Big[2\Big(\frac{\pi}{2} - \phi_0\Big) - \sqrt3\Big]h(\phi_0) > 0,$$

since $2\pi/3 > \sqrt3$. Hence $D_0(\varpi) < 0 < D_0(x_0)$, so $\varpi < x_0$ and $\operatorname{am}\varpi < \phi_0$, and therefore

$$\tilde e = 2E - E(\operatorname{am}\varpi\,|\,m) > 2E - E(\phi_0\,|\,m) > \sqrt3\,w(\phi_0) = \sqrt3\sqrt{1 - m/4} > \sqrt3\cdot\tfrac{\sqrt3}{2} = \tfrac32.$$

(iii) Since $1/w - w = m\sin^2\theta/w$, $K - E = m\int_0^{\pi/2}\sin^2\theta\,w(\theta)^{-1}d\theta \leq m\,\pi/(4\sqrt{1-m})$, so $2(1-m)(K-E) \leq \frac{\pi}{2}\,m\sqrt{1-m}$. The function $m\sqrt{1-m}$ decreases for $m > 2/3$, and $m > k_0^2 > 3/4$ by (i), so $m\sqrt{1-m} < \frac34\cdot\frac12$ and $2(1-m)(K-E) < 3\pi/16 < 3/4$. □

Proposition A.3 ($Q < 0$ on the right sheets)**.** *For every $n \geq 1$ and $0 < k < 1$, $Q < 0$ on the open sheet Σ_n^+. For $k < k_0$ this includes its α_1 end $ab = 1$; the critical end at $k = k_0$ has $Q = 0$.*

Proof. Every term of Q has even total degree in (a,b,c), so Q is invariant under the simultaneous negation of (a,b,c) and we may work with the sign-normalized variables of (23): $a = g + \mathsf{u}$, $b = g - \mathsf{v}$, $c = -g$, $g = |\operatorname{cn}p| \in (0,1)$, $\mathsf{u} = \rho/q$, $\mathsf{v} = \rho/e$, $\rho = |\operatorname{sn}p|\operatorname{dn}p$, on $p = 2nK + \xi$ with $0 < \xi < K$. On the sheet $b > 0$, so $\mathsf{w} := \mathsf{v}/g < 1$; $0 < ab \leq 1$ with equality exactly at the α_1 end; and $e = \mathcal{E}(p) > 2nE > 2$, so $\mathsf{v} < \rho/2$. Put $L_0 = (1-m)/m$, $\mathsf{x} = \mathsf{u}/g$, $\mathsf{t} = L_0\mathsf{v}/\mathsf{u} = (1-m)q/(me)$, $\mathsf{T} = 2\mathsf{t}$ and $\mathsf{y} = g^2$.

Step 1: $\mathsf{t} < 1/2$ *on every cell $n \geq 1$.* The inequality is equivalent to $F(p) = mp - (2-m)q(p) > 0$. By quasi-periodicity $q(p) = 2n(K-E) + \xi - \mathcal{E}(\xi)$, so $F(p) = 2nD + F_0(\xi)$ with

$$D = mK - (2-m)(K-E) = (2-m)E - 2(1-m)K, \qquad F_0(\xi) = m\xi - (2-m)\big(\xi - \mathcal{E}(\xi)\big).$$

Writing $D = m\int_0^{\pi/2} Y(\theta)/w(\theta)\,d\theta$ with $Y = 1-(2-m)\sin^2\theta$ and $w = \sqrt{1-m\sin^2\theta}$, and pairing θ with $\pi/2-\theta$, where Y becomes $m-Y$ and w becomes $B_0 = \sqrt{1-m+m\sin^2\theta}$, the integrand of the pair is $Y/A_0 + (m-Y)/B_0$ with $A_0 = w(\theta)$; on $0 \le \theta \le \pi/4$ one has $Y \ge m/2 > 0$, and when $Y > m$ the identity, with $x = \sin^2\theta \le 1/2$,

$$Y^2B_0^2 - (Y-m)^2A_0^2 = m(2x-1)\{-(1-m) - x(1-x)(2-m)^2\} > 0$$

gives $Y/A_0 > (Y-m)/B_0$. So $D > 0$. On $[0,K]$, $F_0(0) = 0$, $F_0(K) = D > 0$ and $F_0'(\xi) = m\{1-(2-m)\,\mathrm{sn}^2\,\xi\}$ changes sign once, from positive to negative; hence $F_0 \ge 0$ and $F(p) = 2nD + F_0(\xi) > 0$.

Step 2: the margins. In these variables $\mathsf{v} < \rho/2$ and $ab \le 1$ read

$$\mathsf{S}_V = (1-\mathsf{y})(L_0+\mathsf{y}) - 4(1+L_0)\mathsf{w}^2\mathsf{y} > 0, \qquad \mathsf{S}_Z = \mathsf{T} - \mathsf{y}(1-\mathsf{w})(\mathsf{T}+2L_0\mathsf{w}) = 2\mathsf{t}(1-ab) \ge 0.$$

For fixed $(L_0,\mathsf{x},\mathsf{t})$ both cut upper intervals in y: S_V is concave in y, equal to L_0 at $\mathsf{y}=0$ and negative at $\mathsf{y}=1$, and S_Z is linear decreasing in y.

Step 3: concavity. Direct substitution gives

$$-Q = \frac{g^2\mathsf{x}}{L_0^3(1+L_0)}\,P^+(L_0,\mathsf{x},\mathsf{t},\mathsf{y}),$$

where P^+ is a polynomial, quadratic in y, with leading coefficient $(L_0-\mathsf{tx})^2(1+\mathsf{x})\{\mathsf{tx} - L_0(3\mathsf{x}+4)\}$ and value $P^+(L_0,\mathsf{x},\mathsf{t},0) = L_0^3\{L_0(\mathsf{x}+4) - 3\mathsf{tx}\}$. Since $\mathsf{tx} = L_0\mathsf{w} < L_0$, the leading coefficient is negative and the value at $\mathsf{y}=0$ exceeds $L_0^4(1+\mathsf{x}) > 0$. So P^+ is strictly concave in y on the admissible interval and positive at its lower end; it is positive on the whole interval as soon as it is positive at the upper end, which lies on $\mathsf{S}_V = 0$ or on $\mathsf{S}_Z = 0$.

Step 4: the face $\mathsf{S}_Z = 0$. There $ab = 1$. Put $r = \mathsf{y}$, $h = \mathsf{w}$, $w_0 = 1-h \in (0,1)$, so that $b = gw_0$ and $a = (gw_0)^{-1}$, and let $\mathcal{T} = g^2w_0^2\,(-Q)$; explicitly

$$\mathcal{T} = m\,(r-1)(rw_0^2-1)^2 + r^2w_0^4 - 3r^2w_0^3 + 3r^2w_0^2 - rw_0 - r + 1,$$

so $\partial_m\mathcal{T} = (r-1)(rw_0^2-1)^2 < 0$ (note $rw_0^2 = b^2 < ab = 1$). The margin $\mathsf{S}_V \ge 0$ says $m \le M = \{(1-r) - 4rh^2\}/(1-r)^2$. If $M \ge 1$, i.e. $r \le 1-4h^2$ (so $h < 1/2$), then $\mathcal{T}(m) > \mathcal{T}(1) = rw_0(rw_0-1)(rw_0^2-2w_0+1)$, and the last factor increases in r, so $rw_0^2 - 2w_0 + 1 \le (1-4h^2)(1-h)^2 - 2(1-h) + 1 = -h^2(2h-3)(2h-1) < 0$; with $rw_0 < 1$ this gives $\mathcal{T}(1) > 0$. If $M < 1$, then $\mathcal{T}(m) \ge \mathcal{T}(M) = rh\,\mathcal{P}(h,r)/(r-1)$ with

$$\mathcal{P} = -4h^5r^2 + 16h^4r^2 - 24h^3r^2 + 8h^3r + 19h^2r^2 - 19h^2r - 10hr^2 + 16hr - 6h + 3r^2 - 6r + 3,$$

whose leading coefficient in r is $-(h-1)^2(4h^3 - 8h^2 + 4h - 3) > 0$, so $\mathcal{P}$ is strictly convex in r; the conditions $0 < M < 1$ confine r to $\max(0, 1-4h^2) < r < (1+4h^2)^{-1}$, and at the endpoints $\mathcal{P}(h,(1+4h^2)^{-1}) = -4h^4(17h+3)/(1+4h^2)^2 < 0$, $\mathcal{P}(h,1-4h^2) = -4h^4(h-1)(2h-3)(2h-1)(4h^2-4h-1) < 0$ for $h < 1/2$, and $\mathcal{P}(h,0) = 3(1-2h) \le 0$ for $h \ge 1/2$. Hence $\mathcal{P} < 0$ on the open interval, $\mathcal{T}(M) > 0$, and $\mathcal{T} > 0$, i.e. $-Q > 0$, throughout the face.

Step 5: the face $\mathsf{S}_V = 0$. There $\mathsf{v} = \rho/2$. Put $r = \mathsf{y}$, $c = \mathsf{w} = \rho/(2g)$ and $z = ab$. The face equation gives $m = M_0/(1-r)^2$ with $M_0 = 1-r-4rc^2$, and $0 < m < 1$ makes $M_0 > 0$ and $M_1 = r + 4c^2 - 1 > 0$; Step 1 reads $z_0 < z < 1$ with $z_0 = r(1-c)\{M_0 + 2rM_1c\}/M_0$. With (r,c) fixed the cleared $-Q$ is a cubic in z. Its first and third Bernstein control points on $[z_0,1]$ have exact positive rational identities with 133 and 111 terms; its second control point is regularized at the cusp $r=1$, $c=0$ by the two charts $c = h/2$, $r = 1-h^2+h^4u/(1+h^2)$ and $c = (1+h)/2$, $r = u/(h^2+2h+2)$, which encode $M_0, M_1 \ge 0$ and leave the margin

$M_0(1 - z_0) > 0$, and has positive rational identities with 246 and 165 terms in the two charts; its fourth control point is the value at $z = 1$, which lies on $\mathsf{S}_Z = 0$ and is positive by Step 4. The four identities, 655 positive rational terms in all, are frozen in `data/universal_right_sv_certificates.json`, which certifies the strict positivity of the Bernstein control coefficients B_0, B_1 (in both charts) and B_2 of the cleared $-Q$ cubic on $[z_0, 1]$ under the margins $M_0, M_1, M_0(1 - z_0) > 0$; the verifier `scripts/universal_right_fold_proof.py` reconstructs them over $\mathbb{Q}$ without an optimizer and also checks the identities of Steps 1, 3 and 4. Hence $-Q > 0$ on the face, and by Step 3 on the whole sheet, the α_1 end included for $k < k_0$. □

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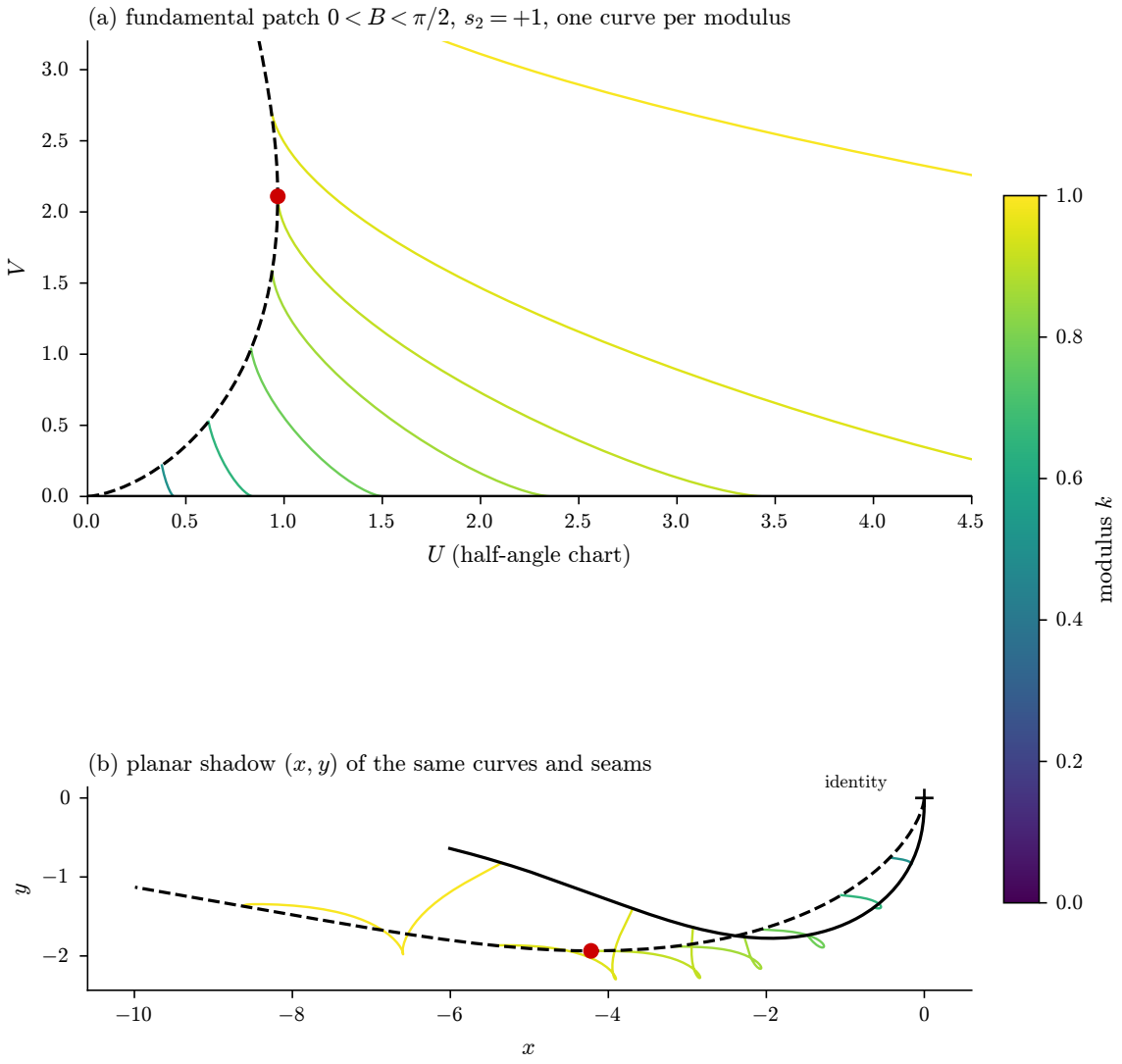


Figure 2: The first caustic. (a) The fundamental patch $0 < B < \pi/2$, $s_2 = +1$, in the half-angle chart (U, V) for seven moduli (color), with the Maxwell seam $B = 0$ (solid, the cusp edge, $V = 0$) and the upper seam $B = \pi/2$ (dashed) across all moduli, and the D_4^+ point at k_0. (b) The planar shadow (x, y) of the same curves and seams; each curve is the projection of the first conjugate points of the geodesics of one modulus.

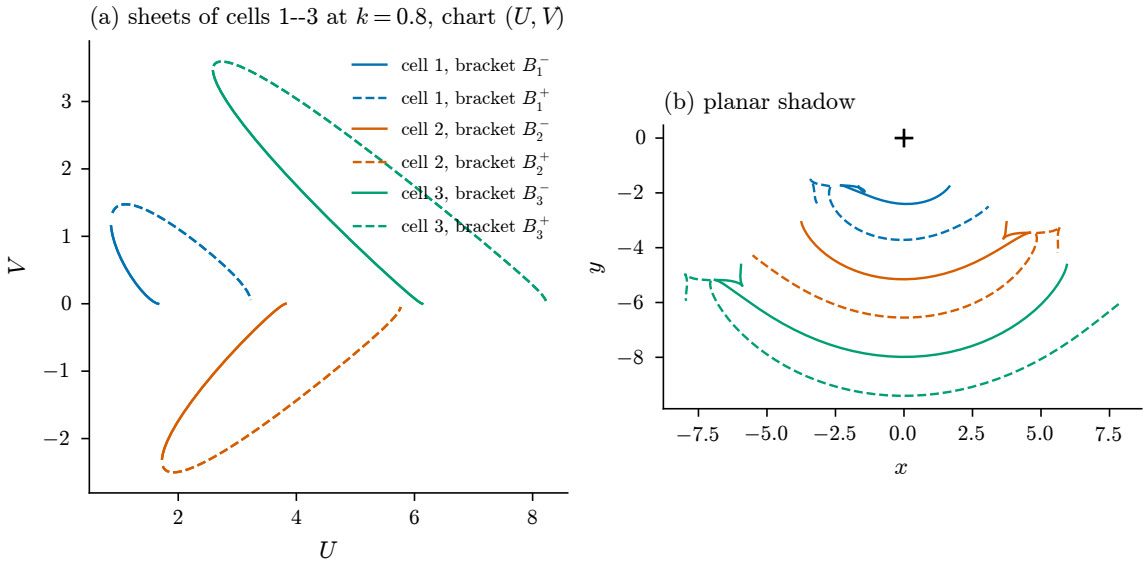


Figure 3: The sheets of the first three cells at $k = 0.8$, both mirror branches: (a) in the half-angle chart, solid for B_n^- and dashed for B_n^+; (b) their planar shadow. Each bracket sheet is a fold surface; its $S = 0$ seam is a cusp edge for B_n^- and a fold for B_n^+, and its $C = 0$ seam is the cusp edge at $2nK$ or the fold at p_n^α (Theorem 9.1).

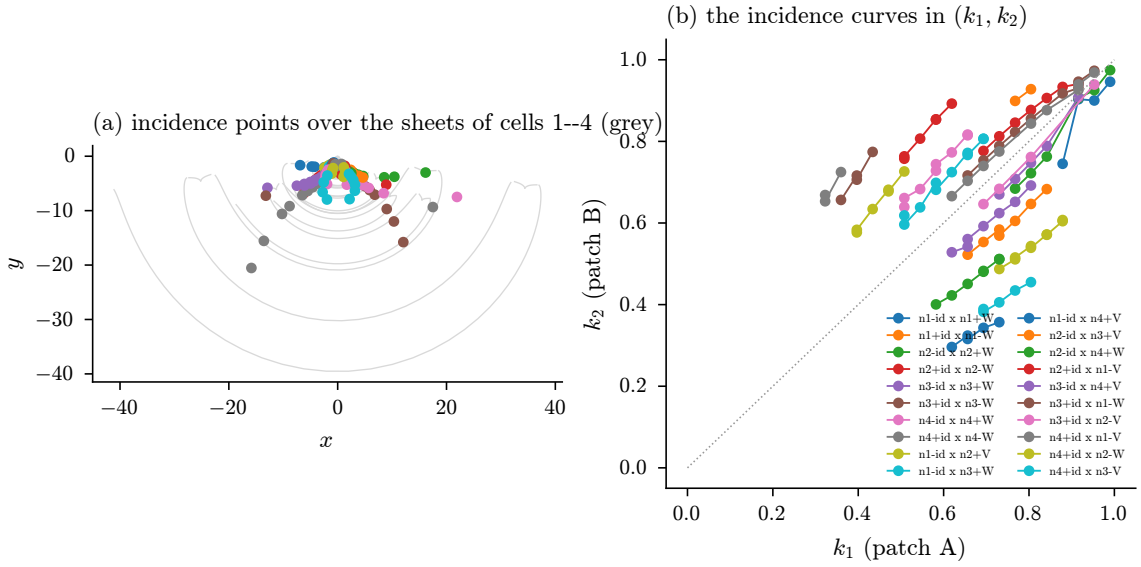


Figure 4: Incidences between the sheets of cells 1–4. (a) The accepted intersection points in the planar shadow, over the sheets in grey; (b) the same points in the (k_1, k_2) plane of the moduli of the two patches, in the order of the label (`patch A`, `patch B` on the axes); each unordered pair appears twice, mirrored in the diagonal, and each color serves two pairs (see the legend). The sampled traces are monotone, and the trace with the left sheet first lies below the diagonal: the left sheet has the larger modulus at these points. Labels: `n1-id x n2+V` is $F(1,-) \cap gF(2,+)$; the token after each cell index and bracket sign names the element of G applied to that patch, `id` the identity (the fundamental patch itself), `V` the reflection $V \mapsto -V$ and `W` the reflection $W \mapsto -W$.